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Su et al.

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(54) **IMAGING LENS DRIVING MODULE AND ELECTRONIC DEVICE**

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G02B 7/02	(2021.01)
H01F 5/04	(2006.01)
H04N 23/51	(2023.01)
H04N 23/55	(2023.01)

(52) **U.S. Cl.**

CPC ***G02B 7/08*** (2013.01); ***G02B 7/026*** (2013.01); ***H01F 5/04*** (2013.01); ***H04N 23/51*** (2023.01); ***H04N 23/55*** (2023.01)

(58) **Field of Classification Search**

CPC G02B 27/646; G02B 7/08; G02B 7/09; G02B 7/04; G02B 27/648; H04N 23/55; H04N 23/57; H04N 23/54; H04N 23/687; H04N 23/51

See application file for complete search history.

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Primary Examiner — Ephrem Z Mebrahtu

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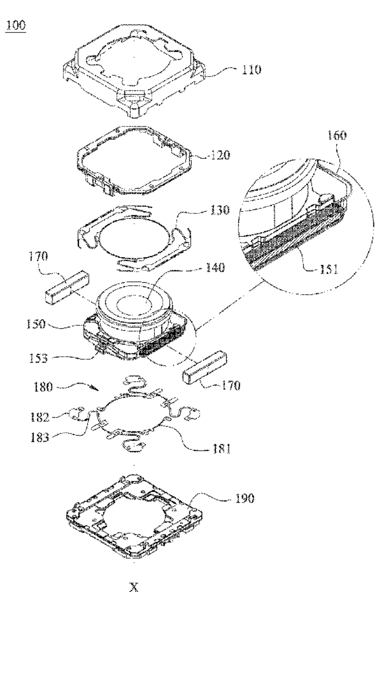
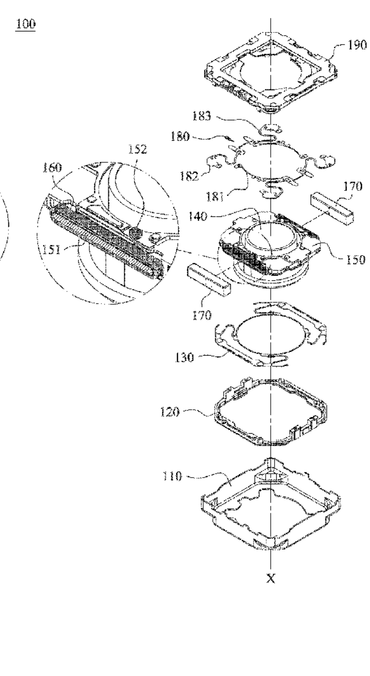
(57) **ABSTRACT**

An imaging lens driving module includes an imaging lens set, a carrier element and a driving mechanism. The imaging lens set has an optical axis. The carrier element is configured to dispose the imaging lens set, and includes an assembling structure. The assembling structure is disposed on an outer surface of the carrier element, and extends along a direction away from the optical axis. The driving mechanism is configured to drive the carrier element to move, and includes at least one coil pair and at least two magnets. The coil pair is disposed on the assembling structure, and includes a bottom layer coil and a top layer coil. The bottom layer coil is wound around and directly contacted with the assembling structure. The top layer coil is stacked on and wound around the bottom layer coil. The magnets correspond to the coil pair, respectively.

5 Claims, 34 Drawing Sheets





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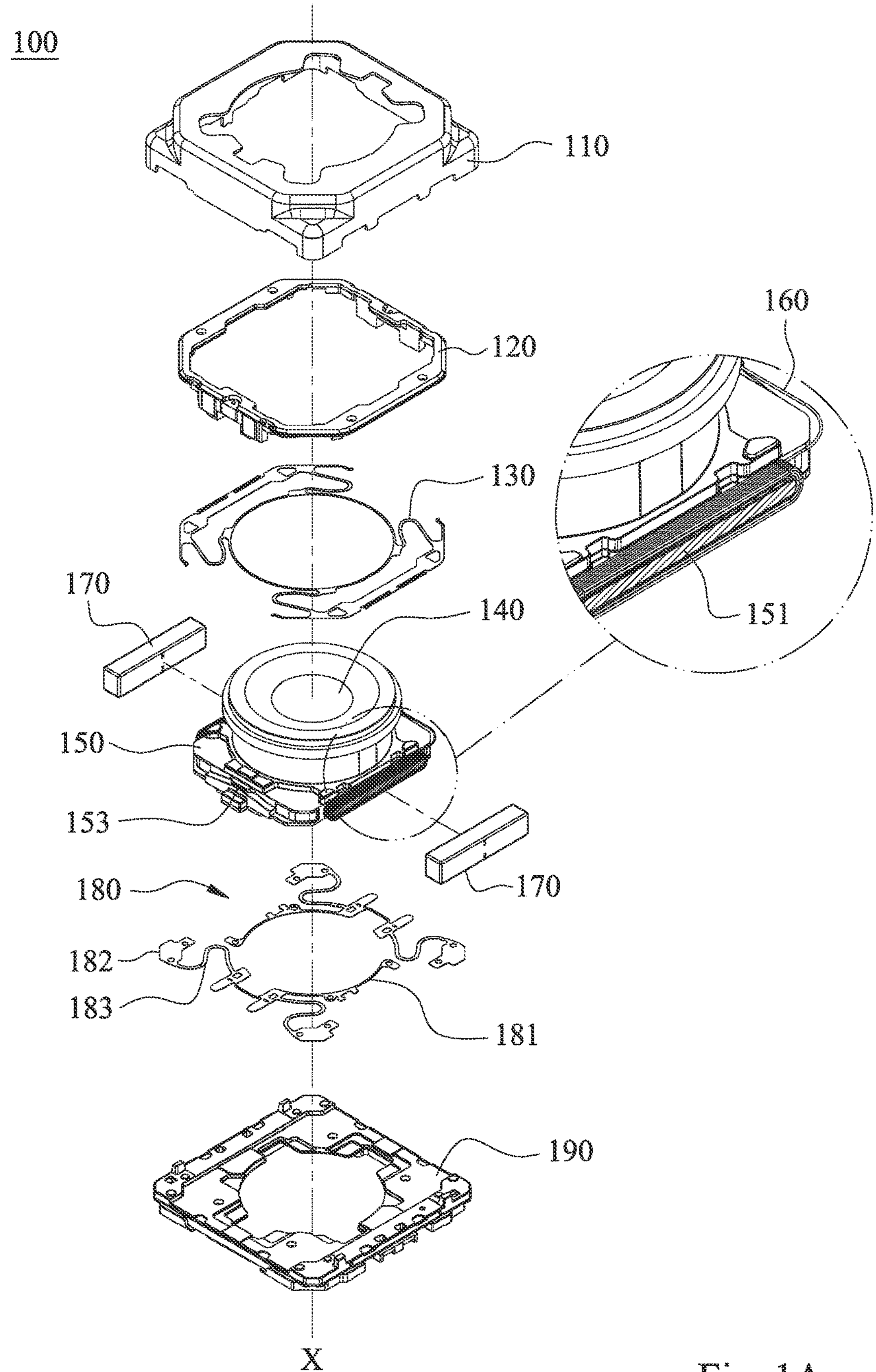


Fig. 1A

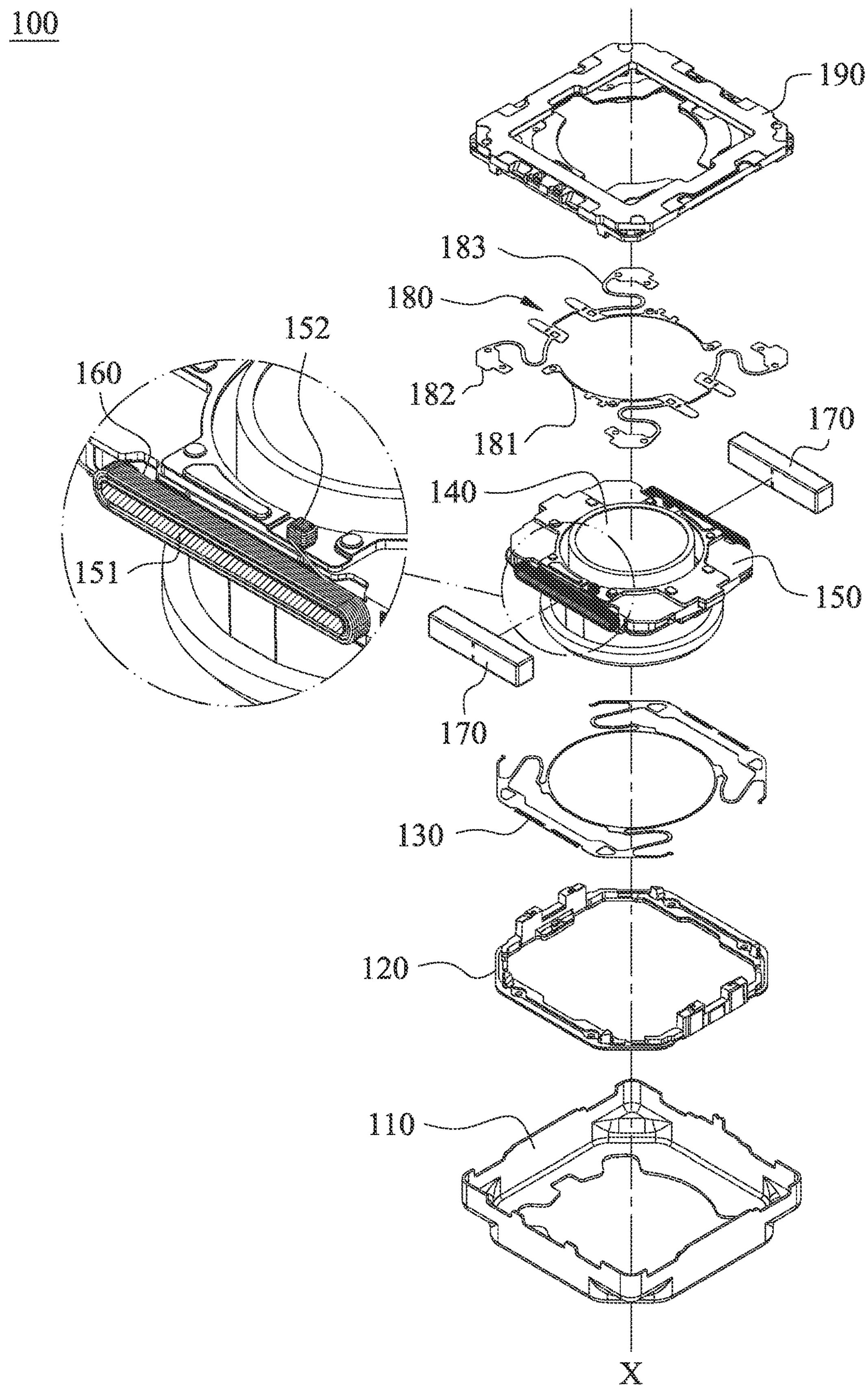


Fig. 1B

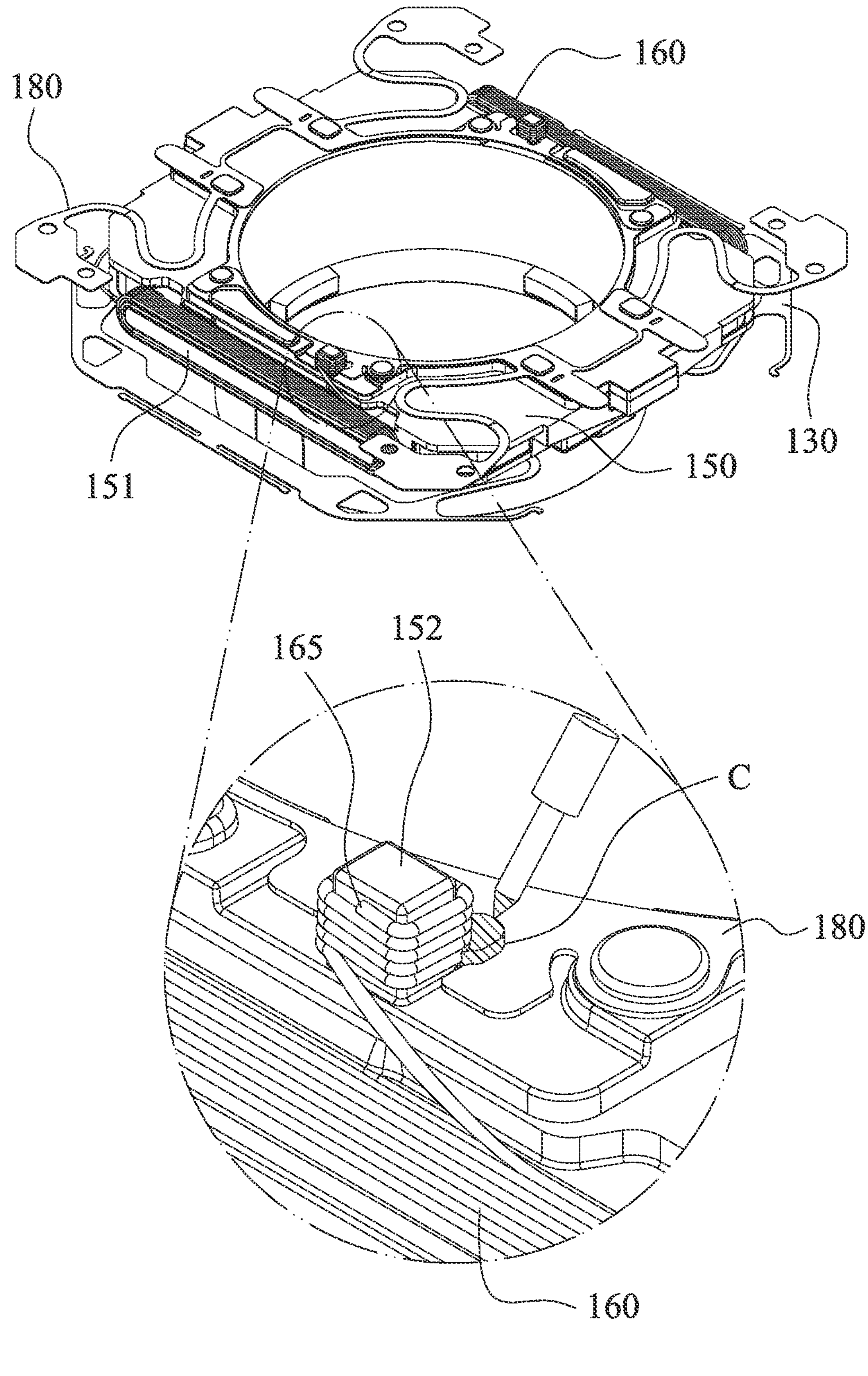


Fig. 1C

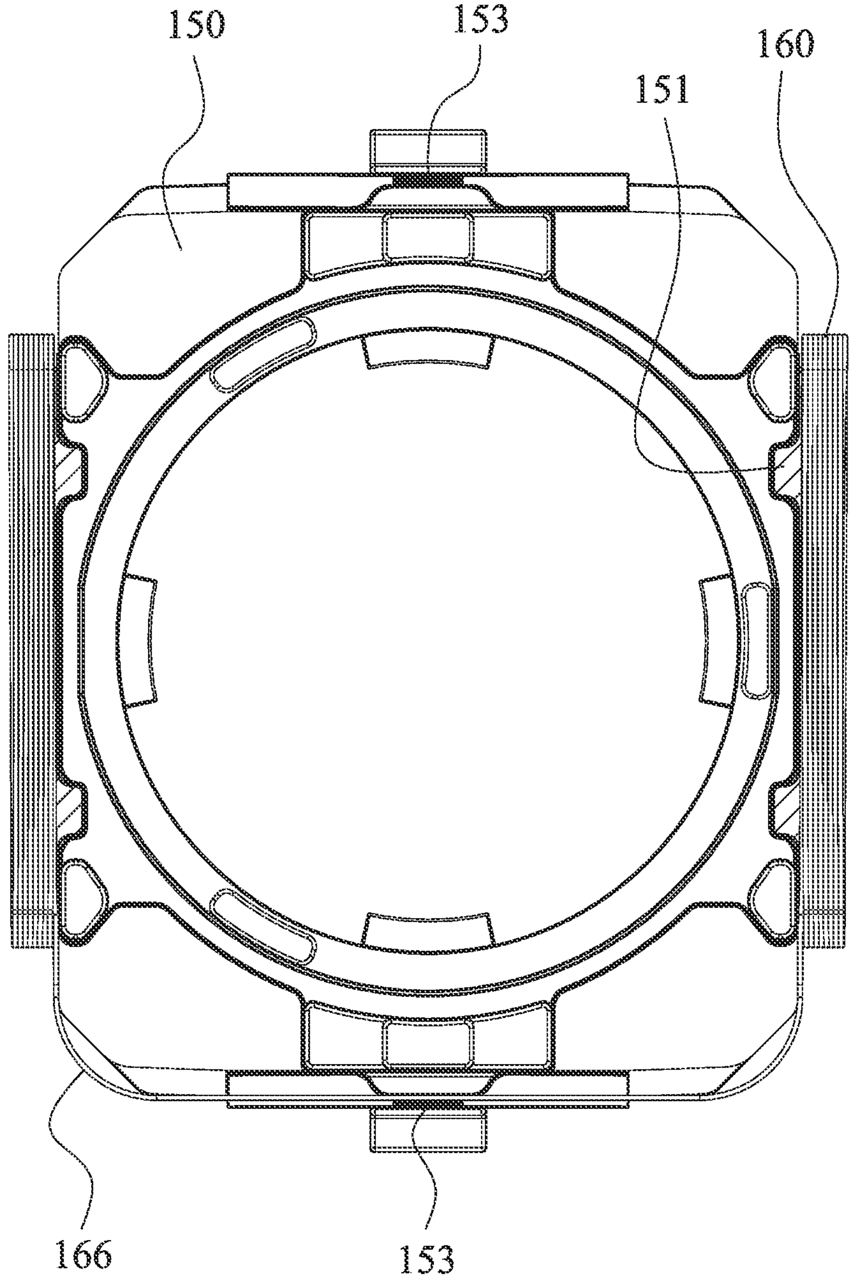


Fig. 1D

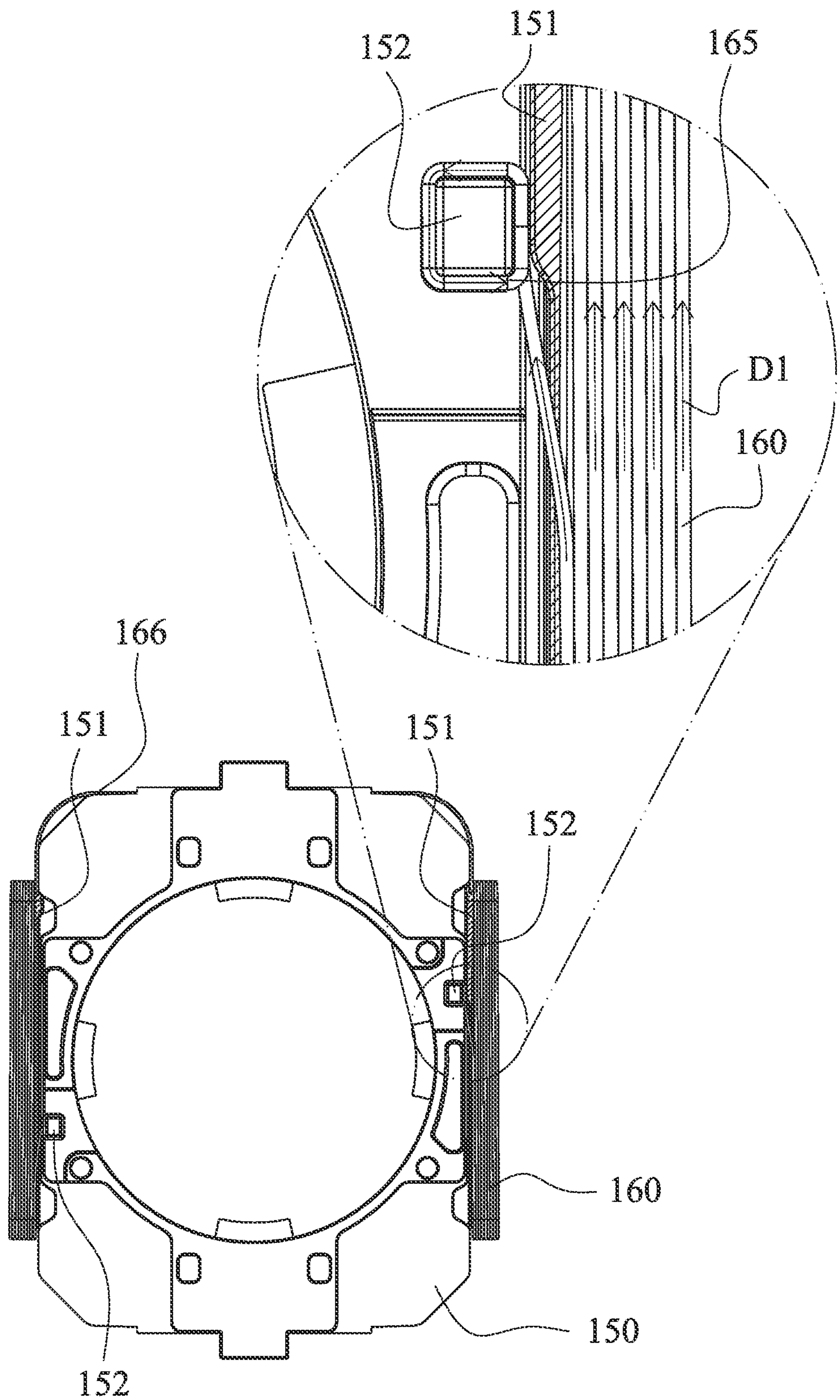


Fig. 1E

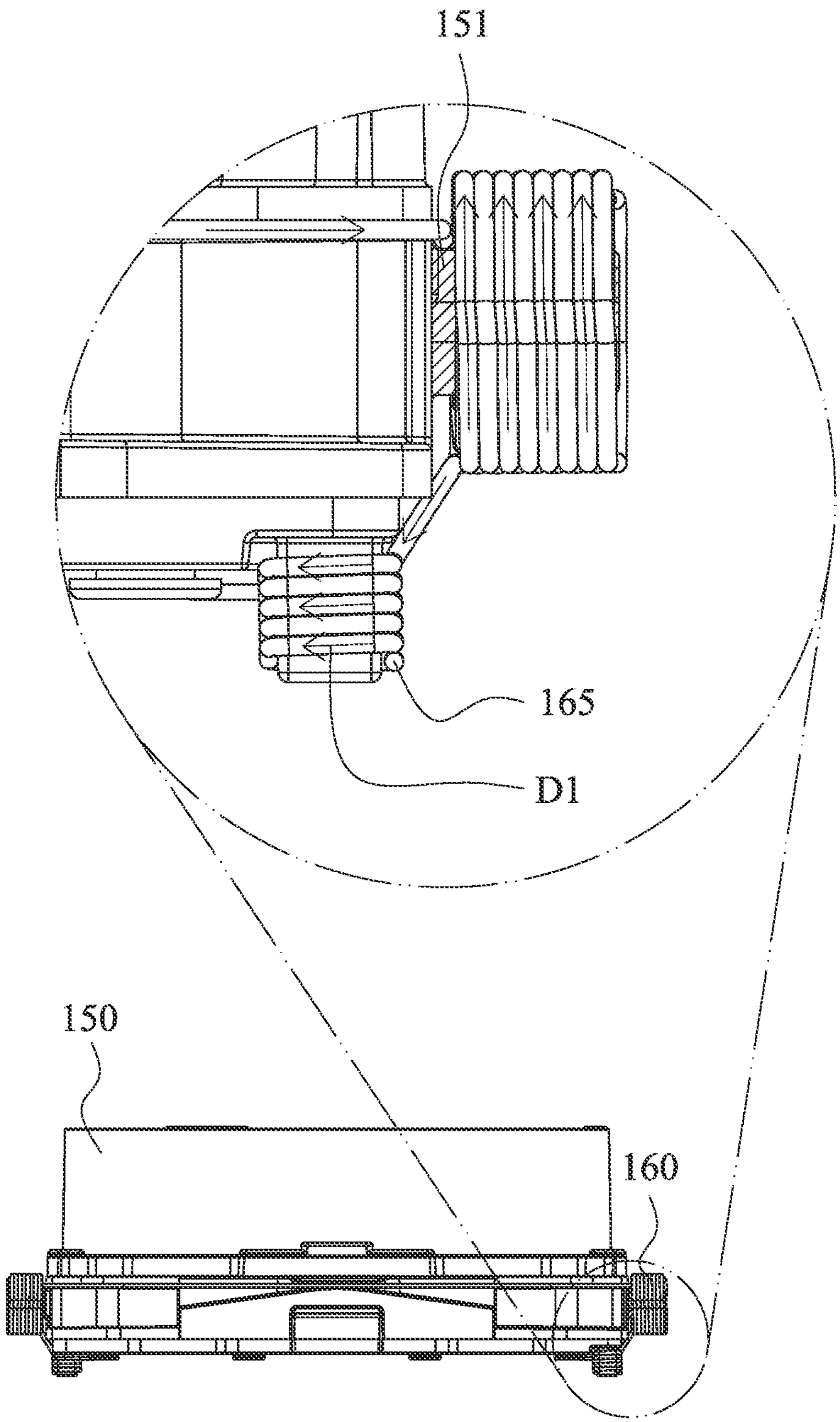


Fig. 1F

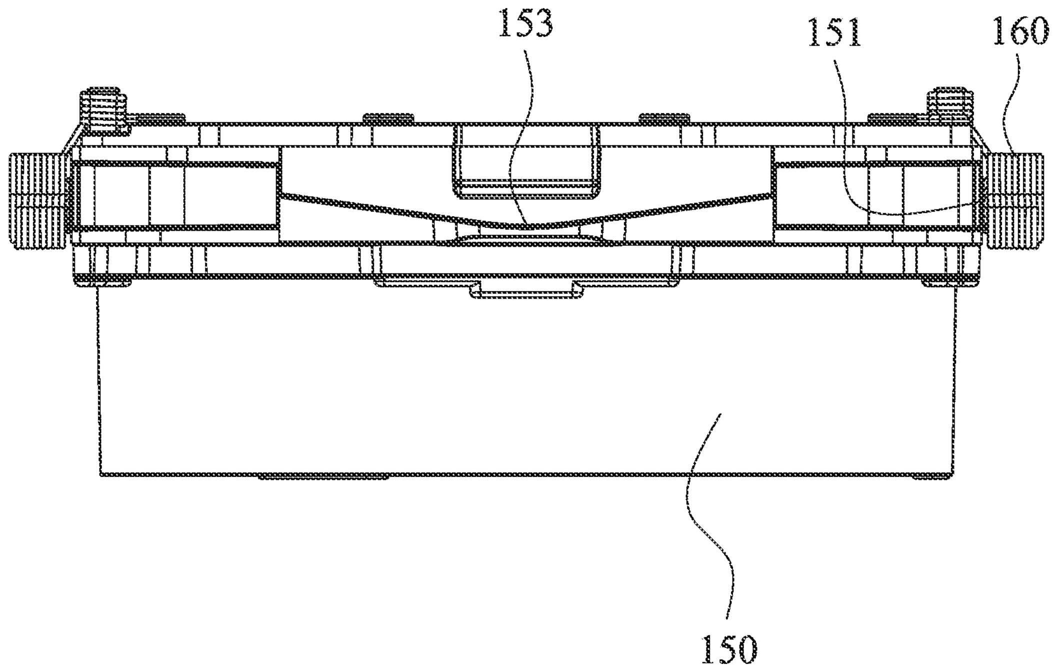


Fig. 1G

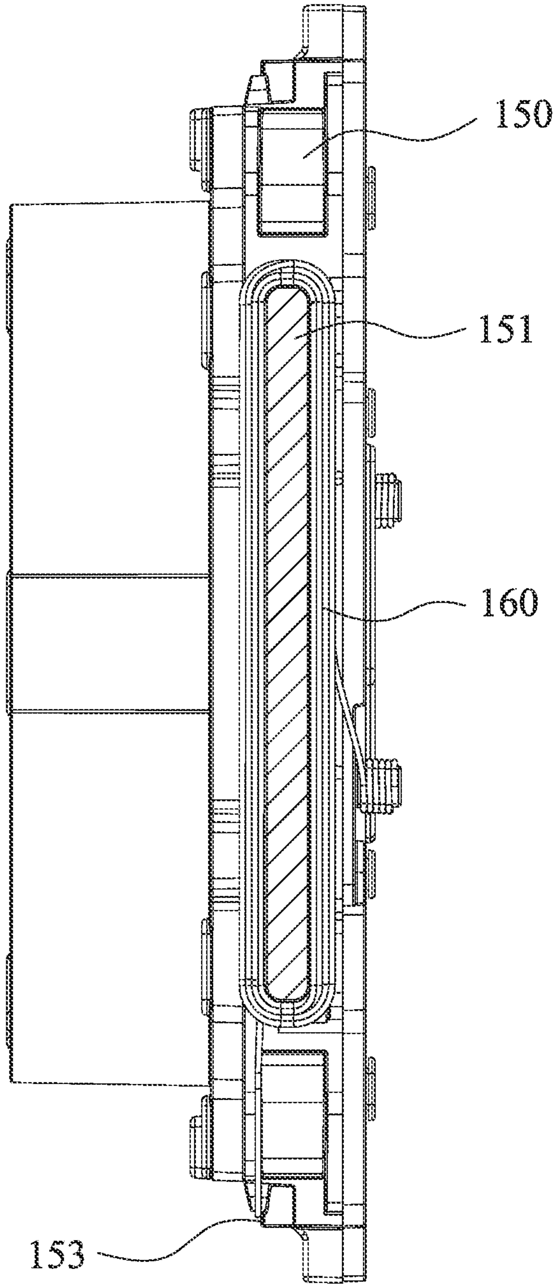


Fig. 1H

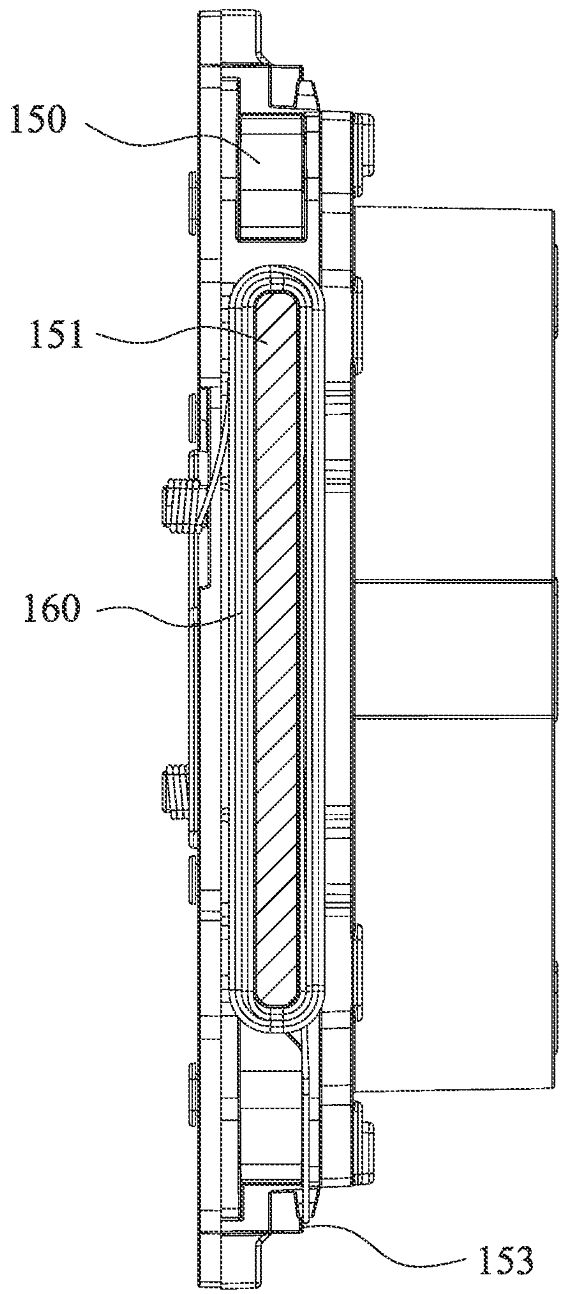
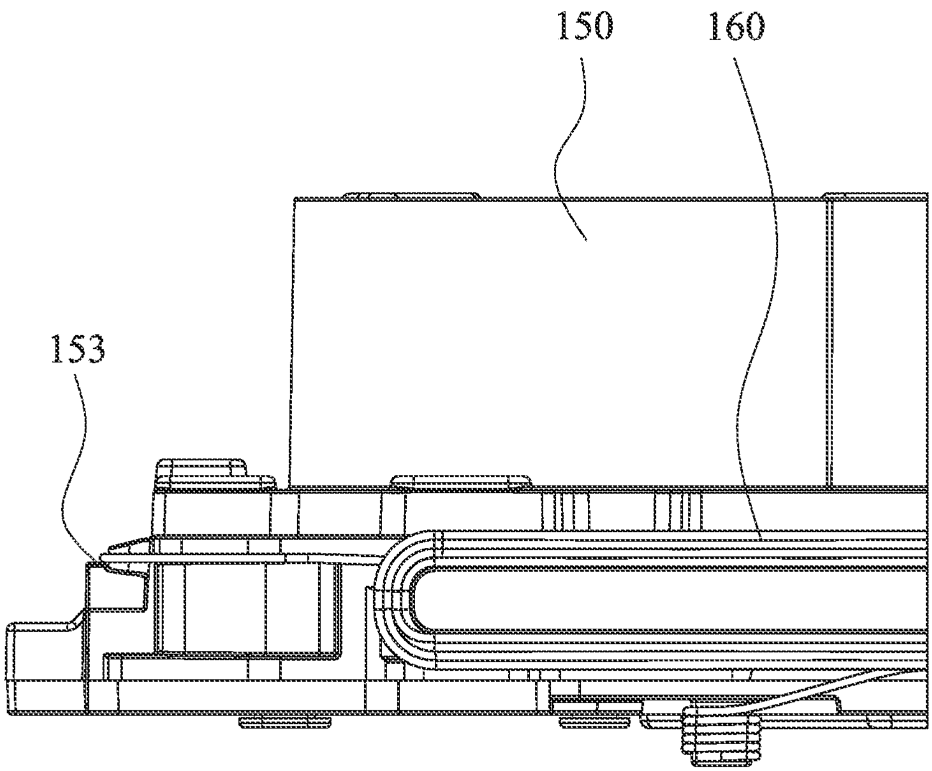


Fig. 1I



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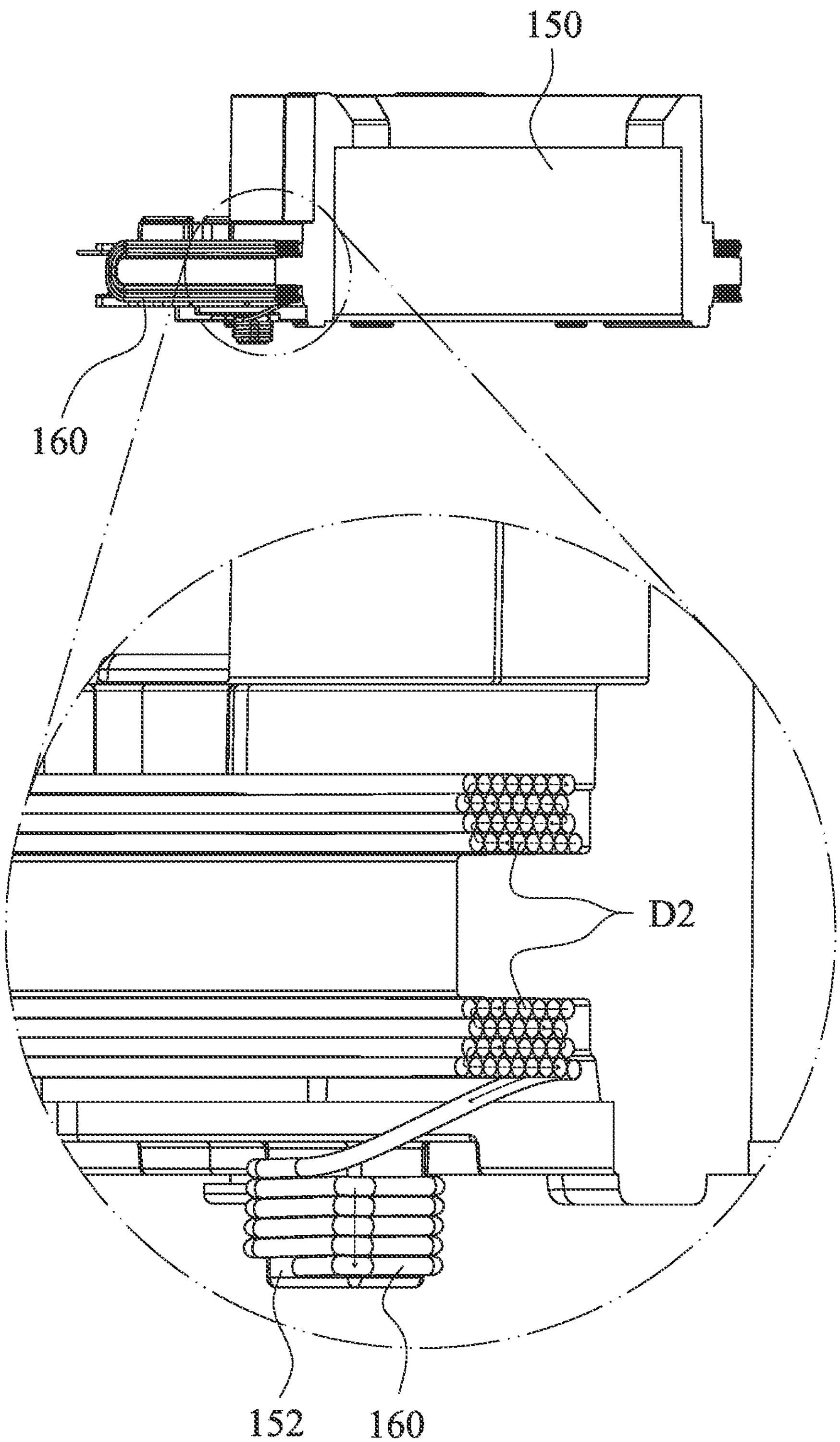

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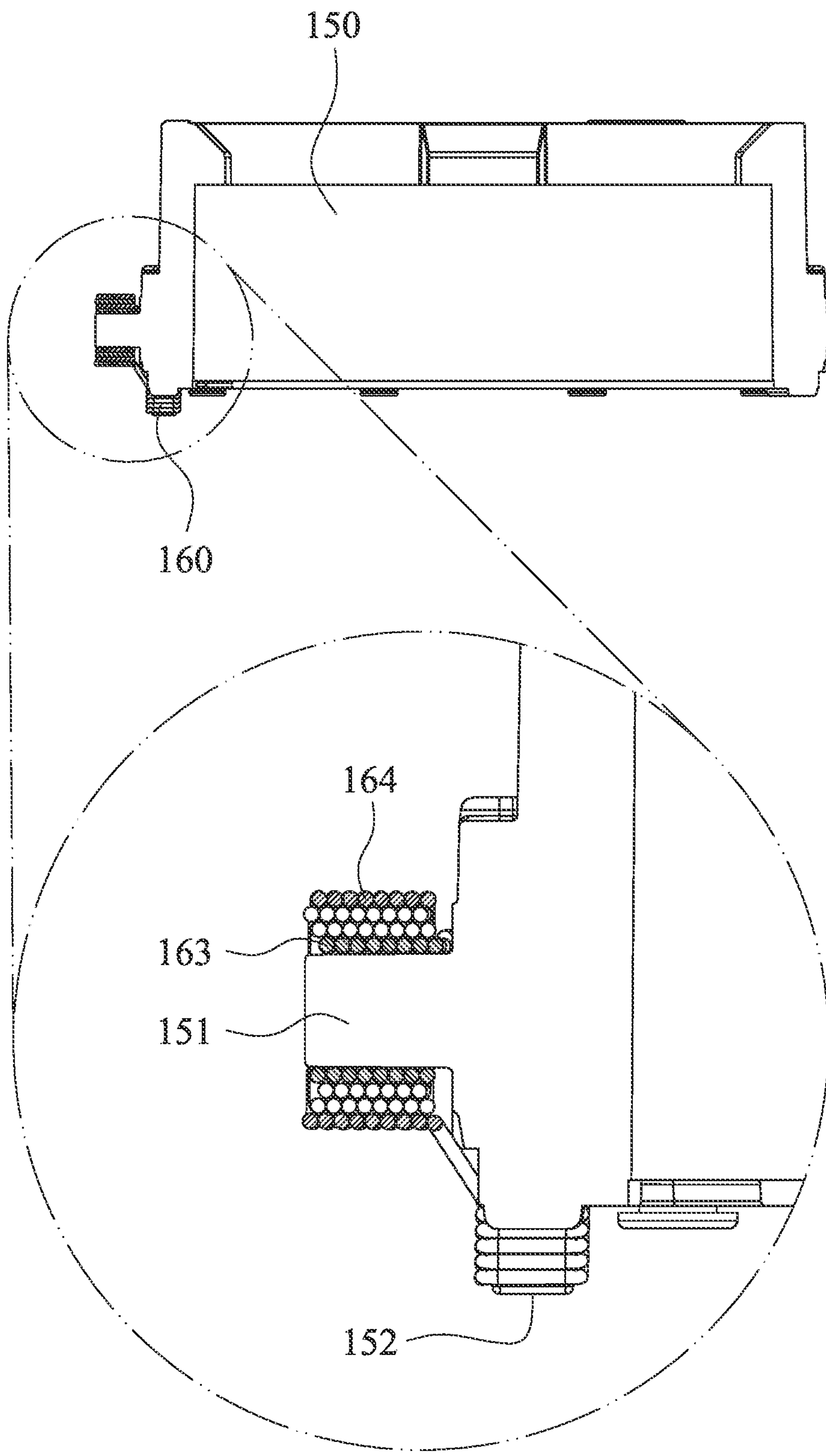

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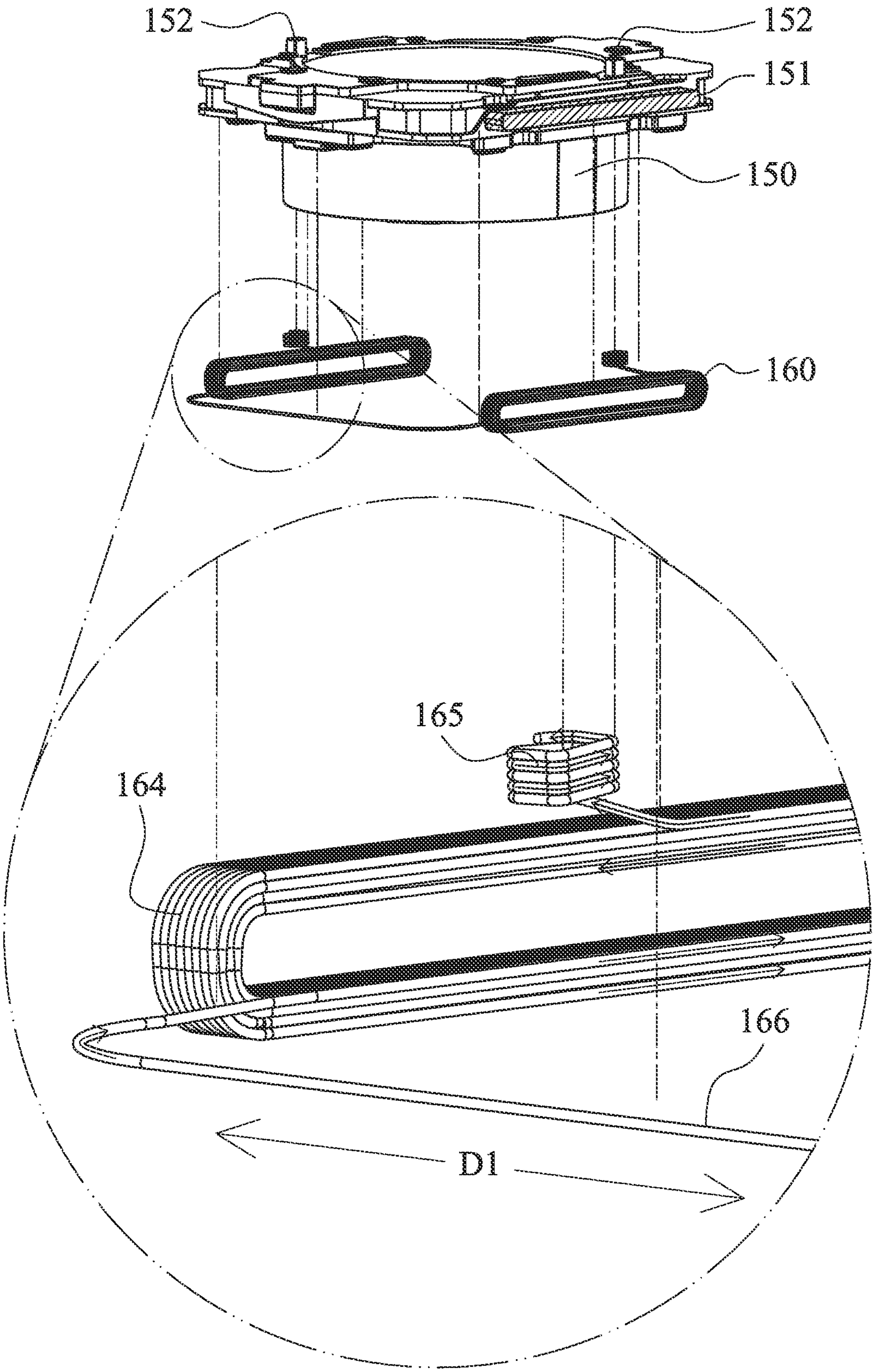

Fig. 1M

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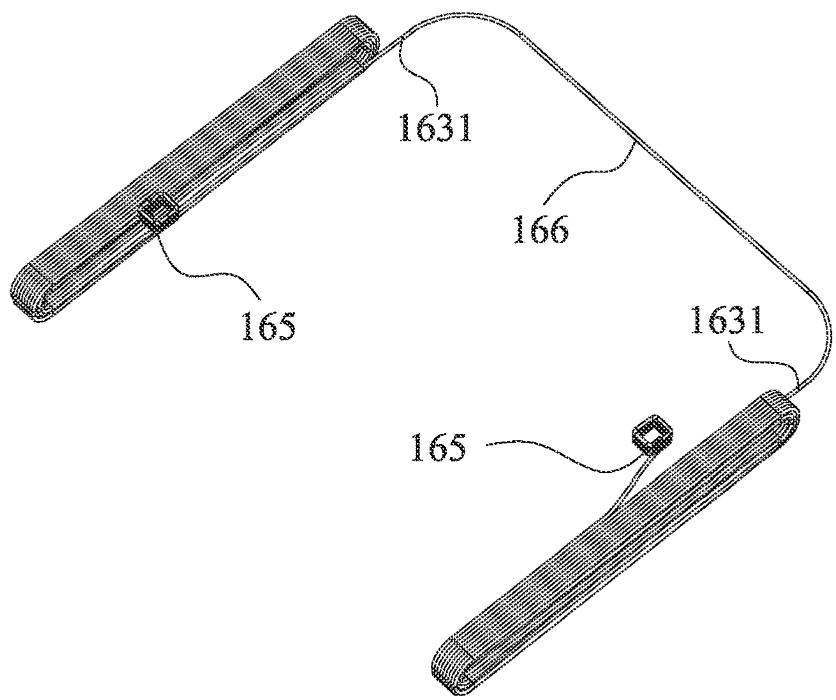

Fig. 1N

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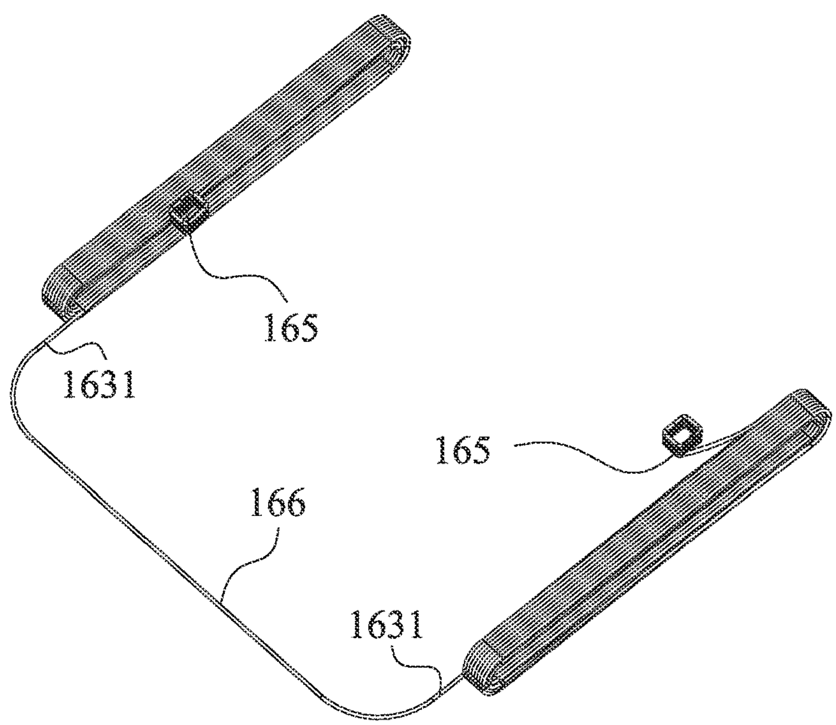

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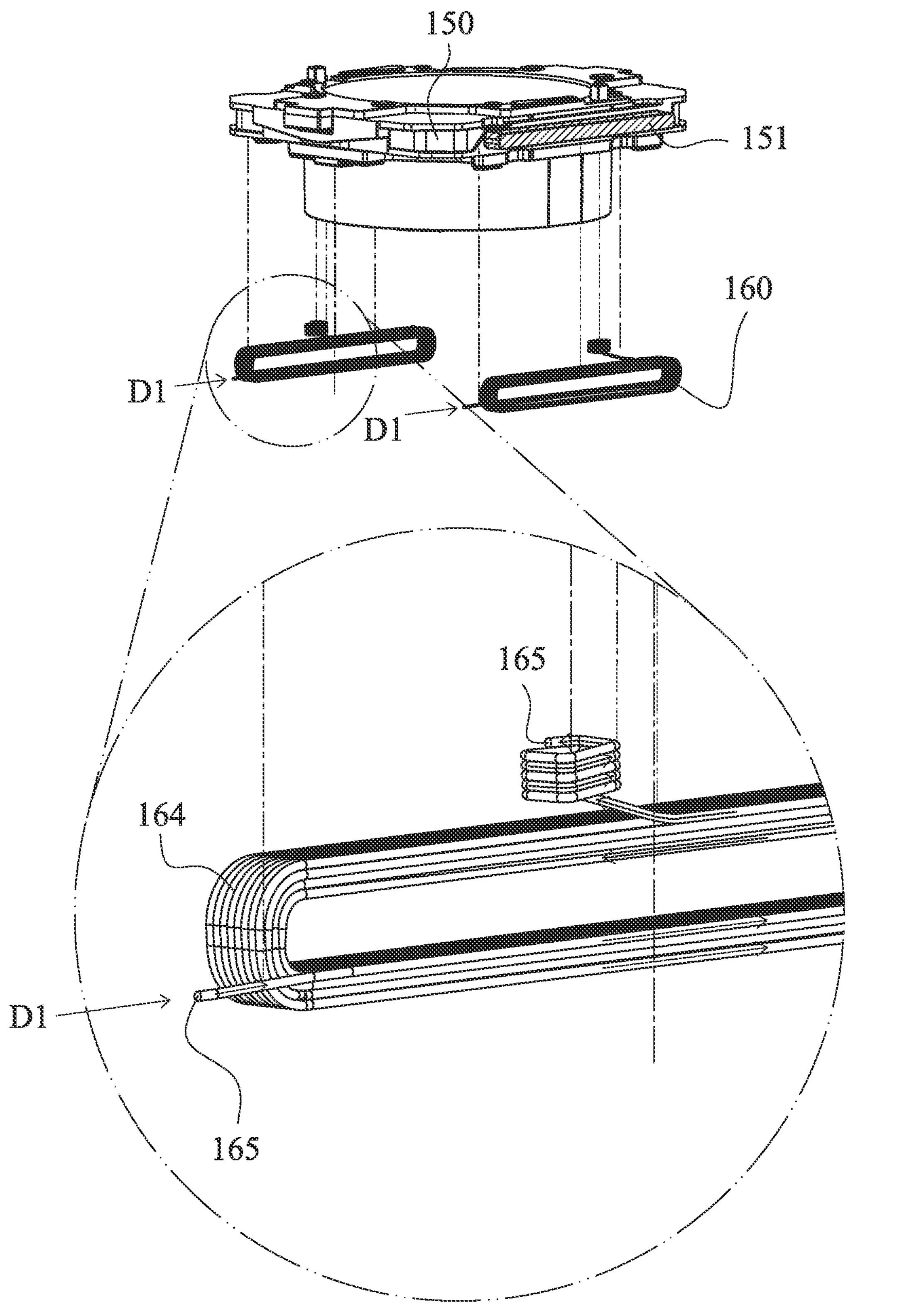

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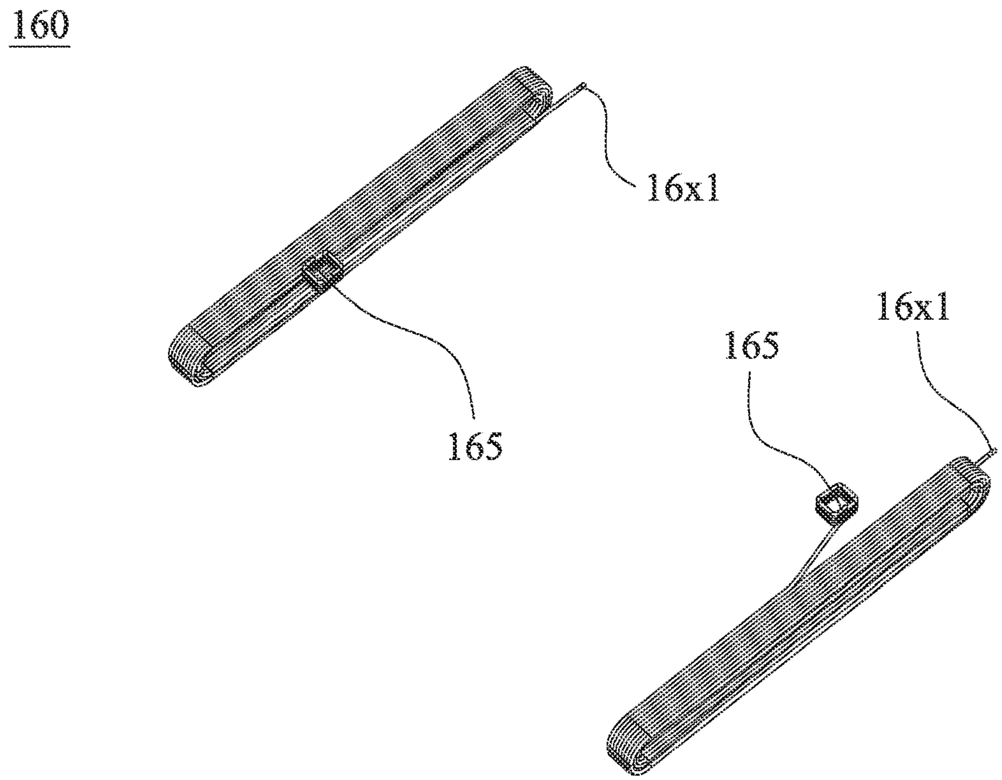

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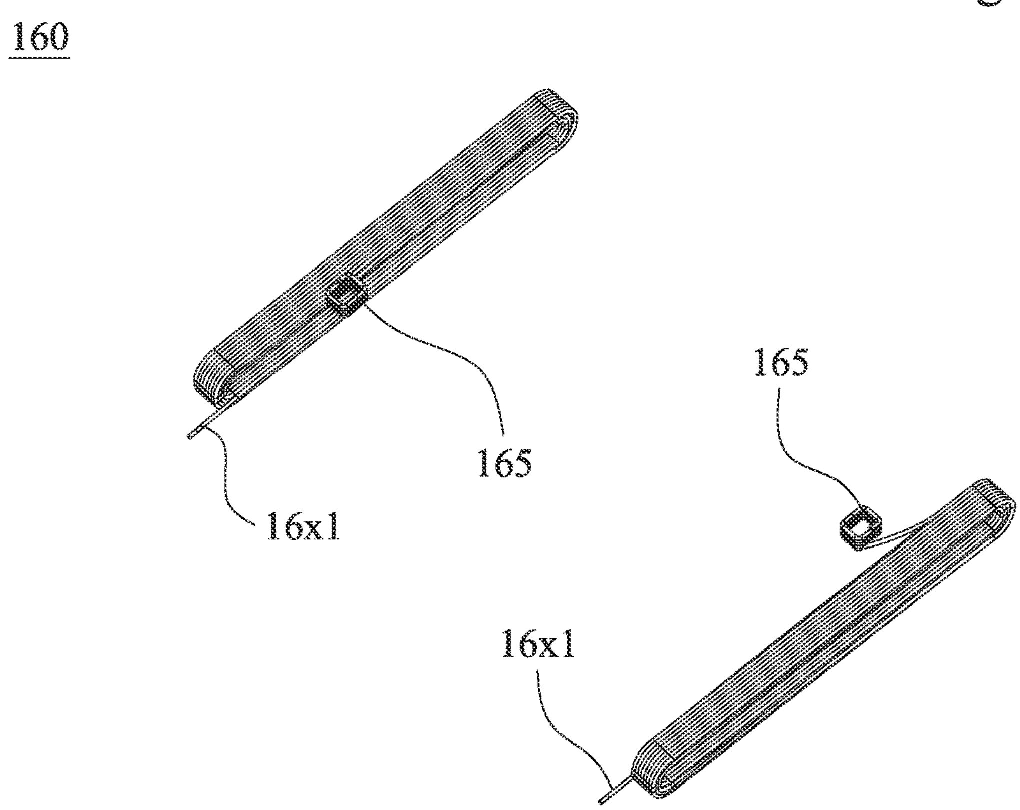

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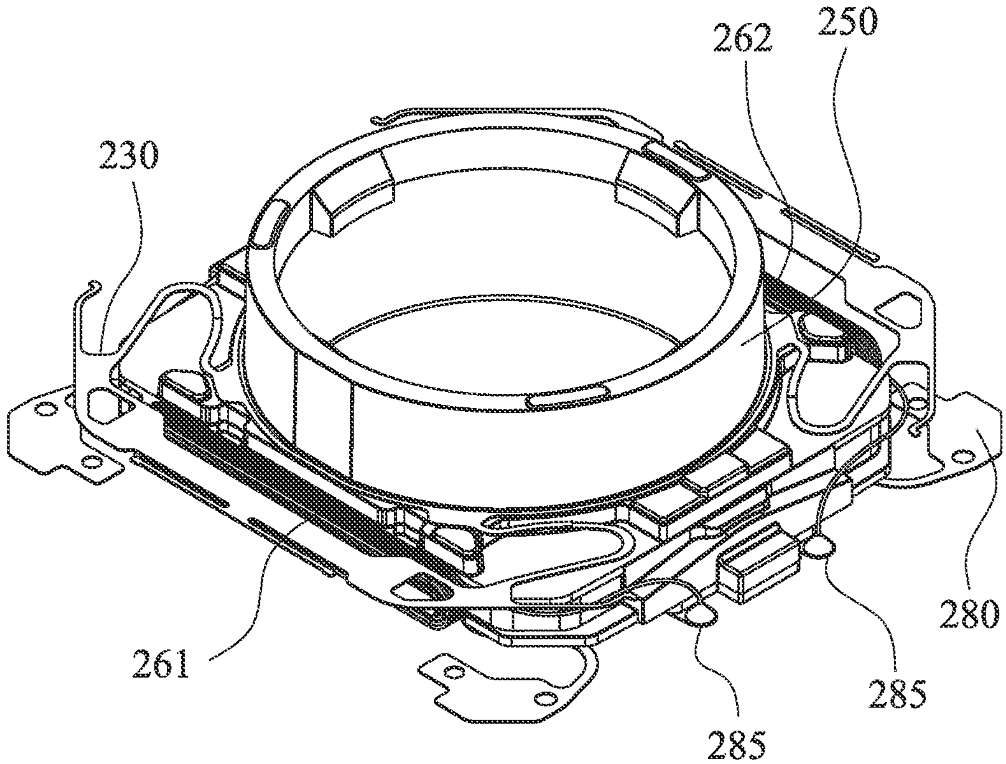

Fig. 2A

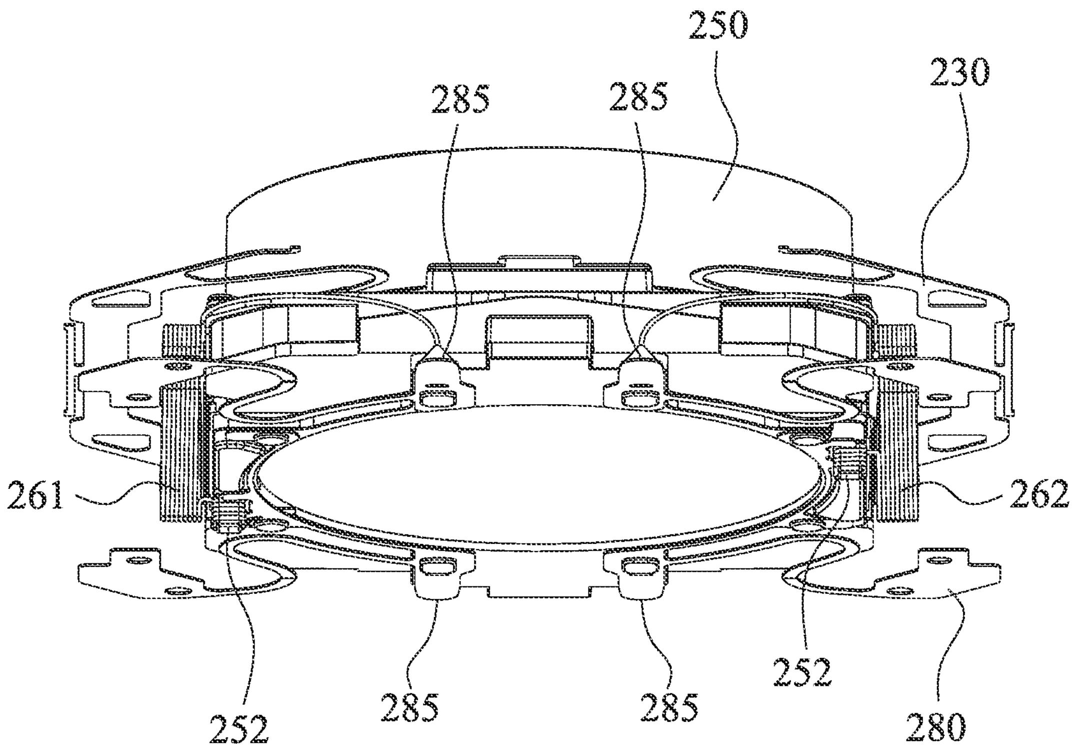
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Fig. 2B

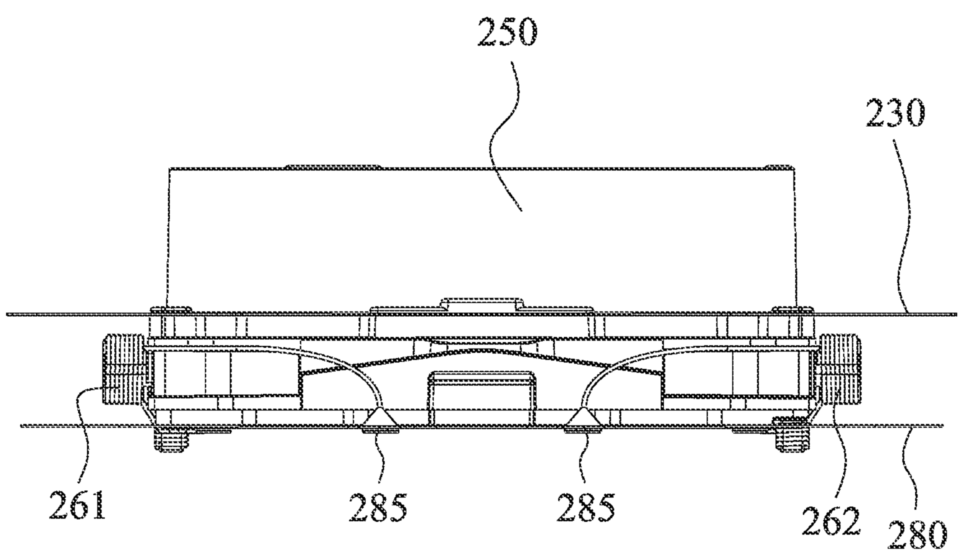
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Fig. 2C

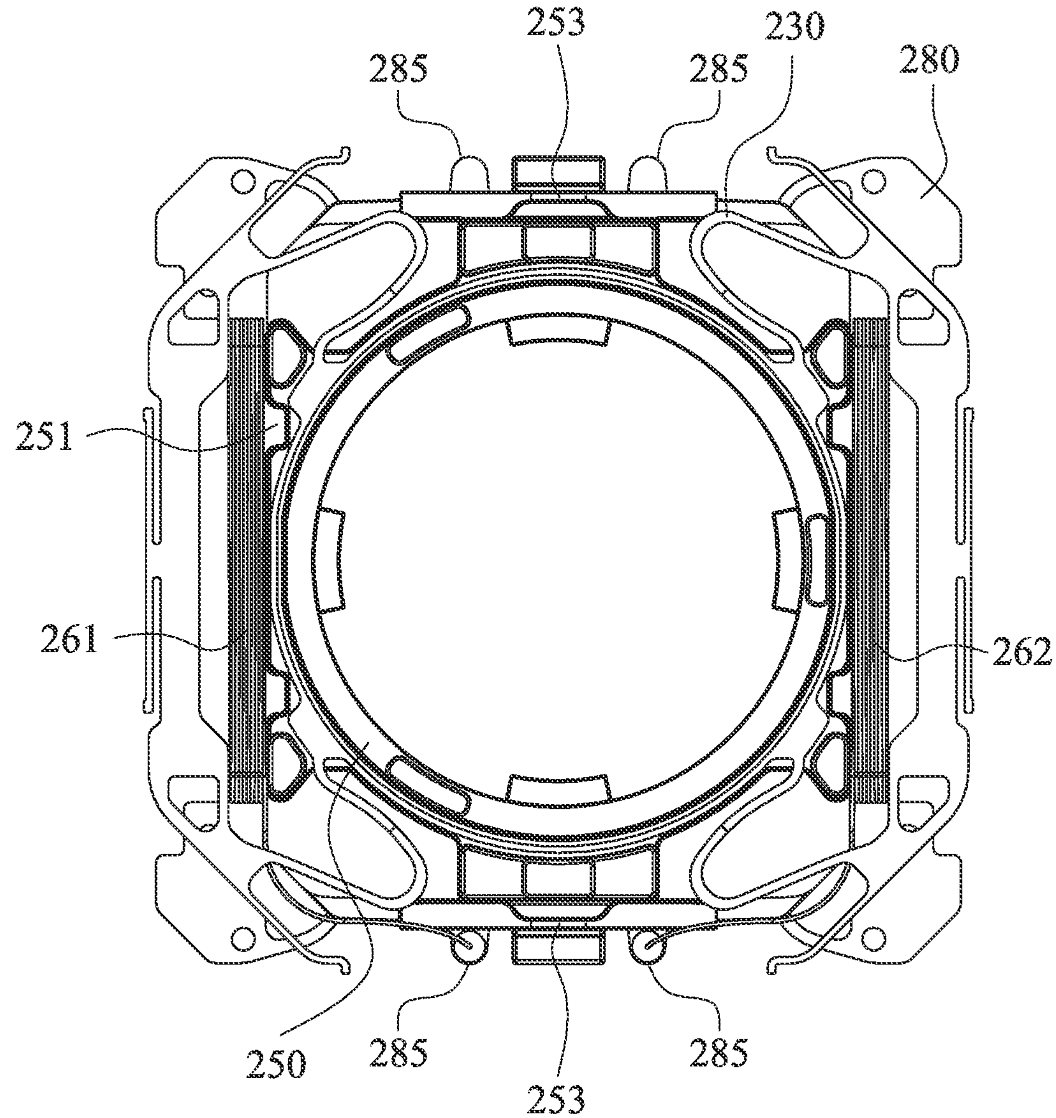
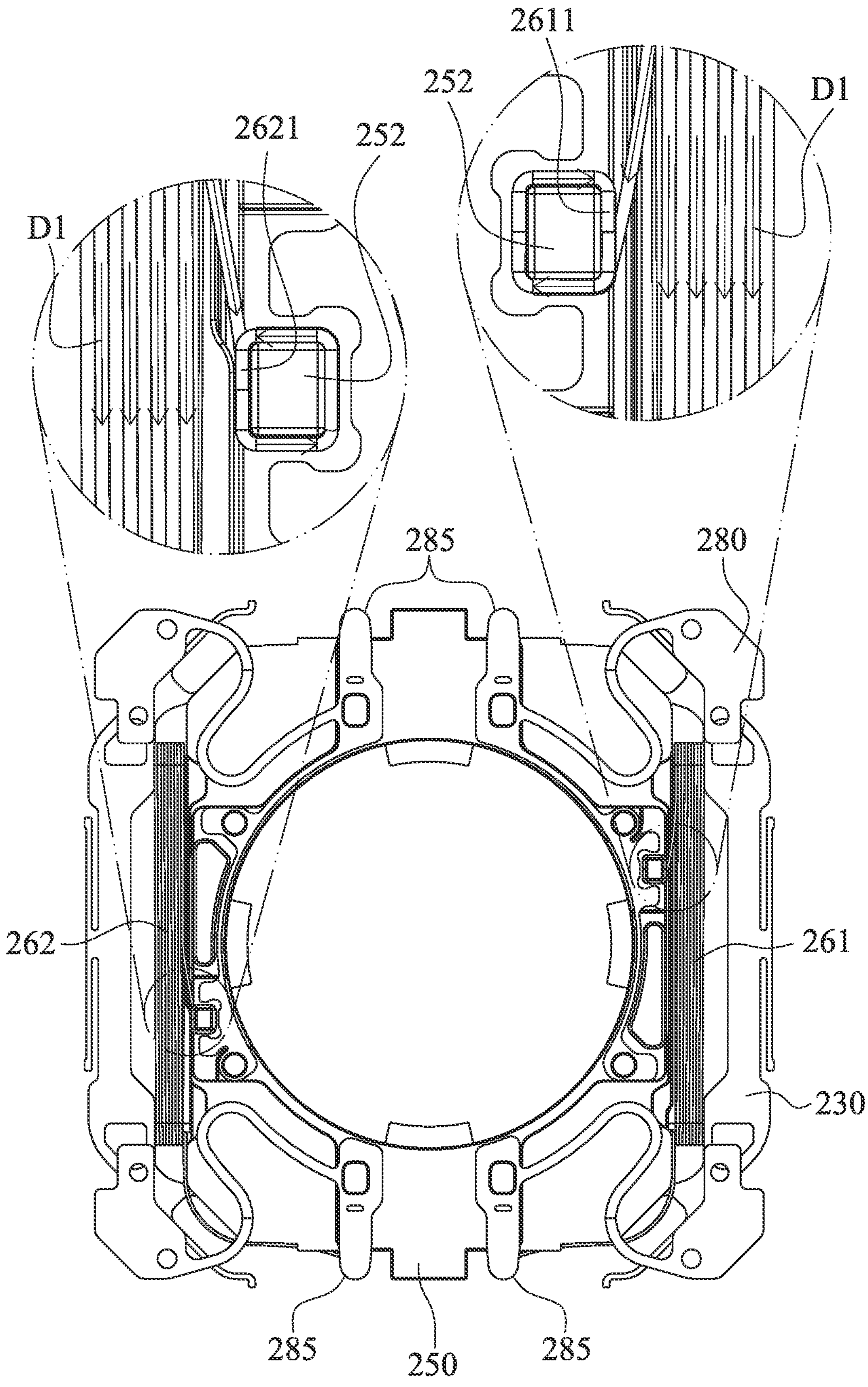


Fig. 2D



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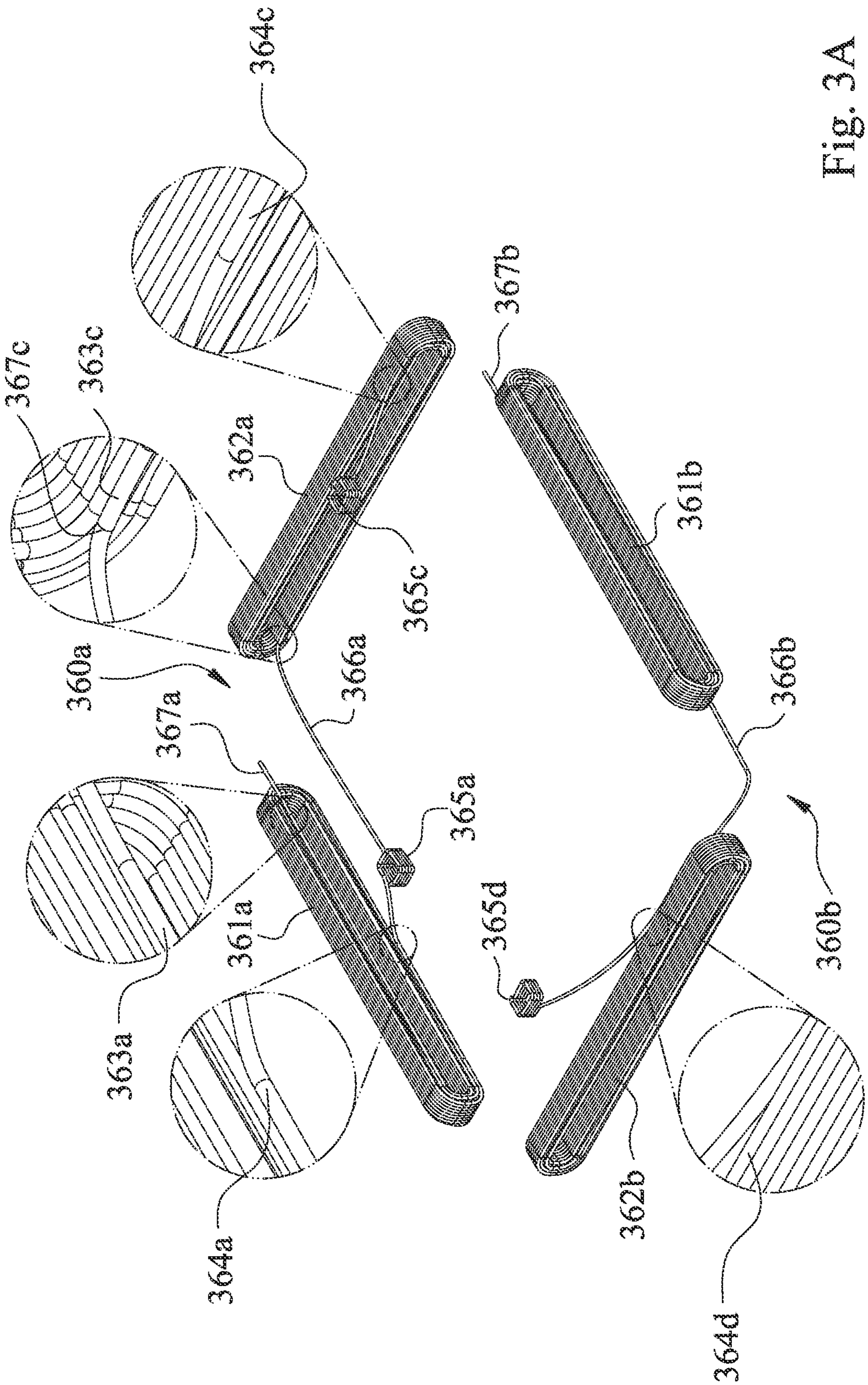
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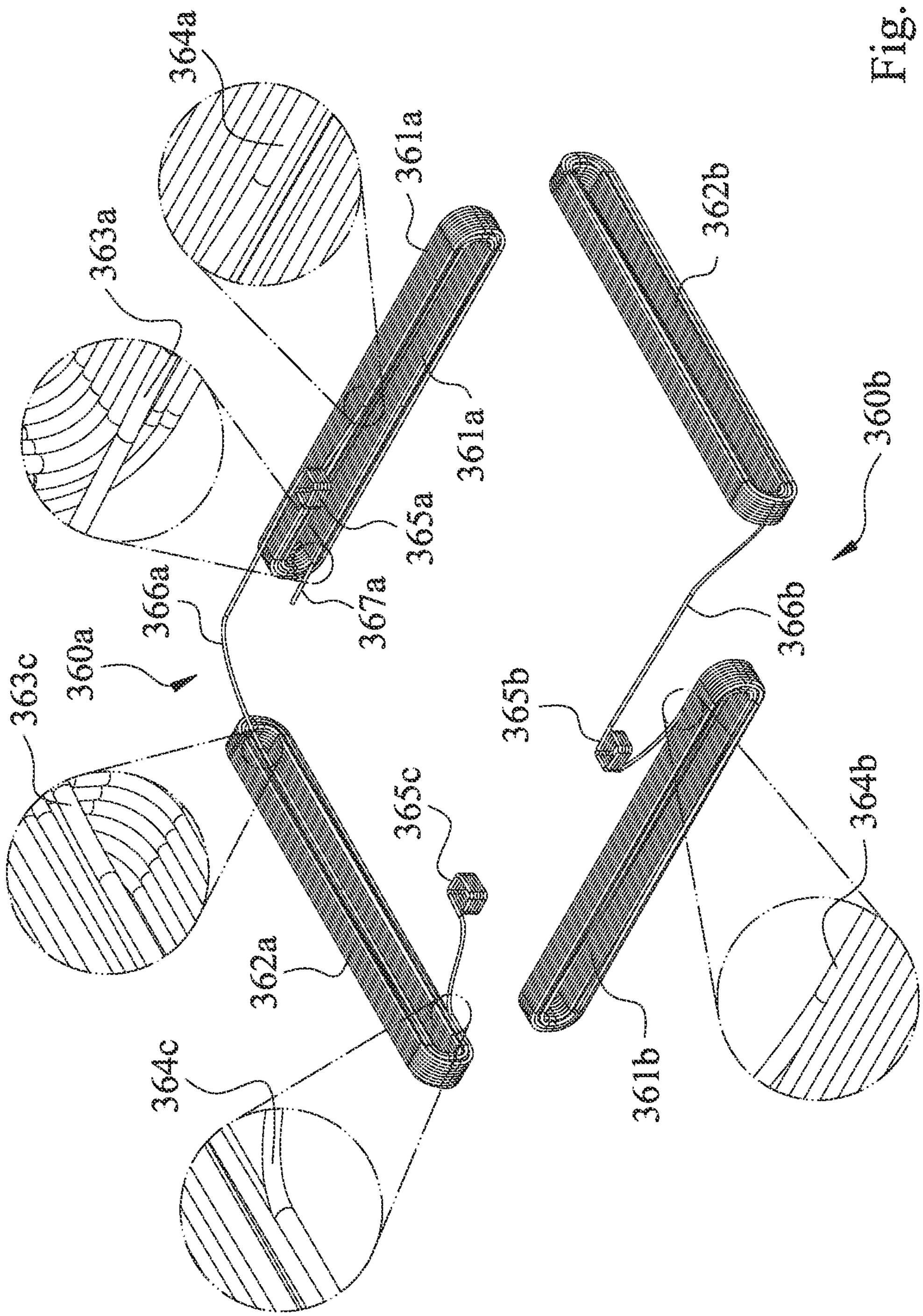
364d

Fig. 3A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361b

Fig. 3B

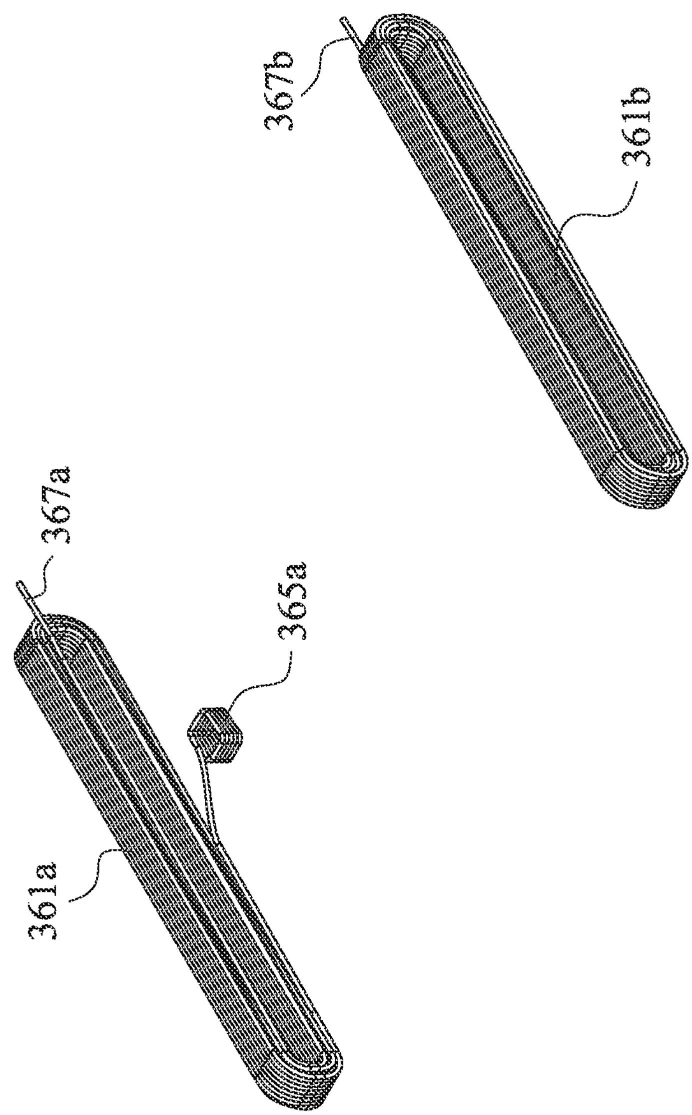

Fig. 3C

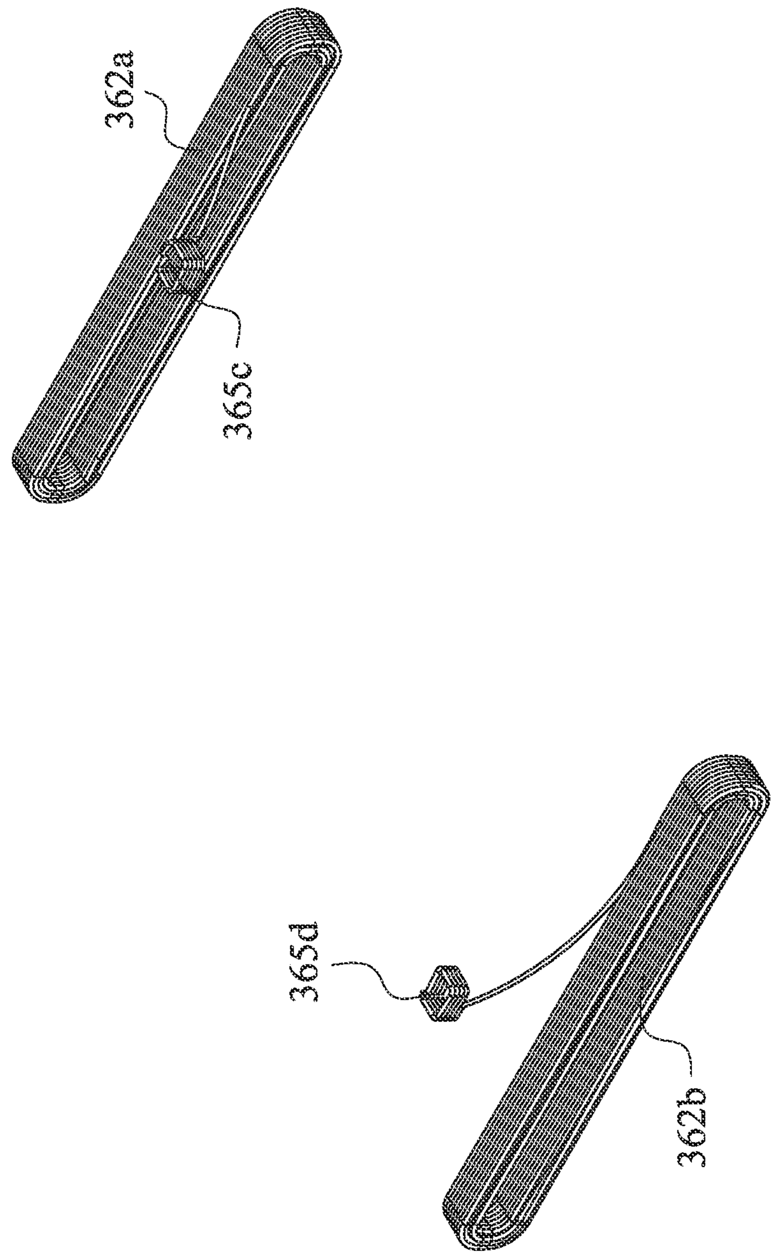

Fig. 3D

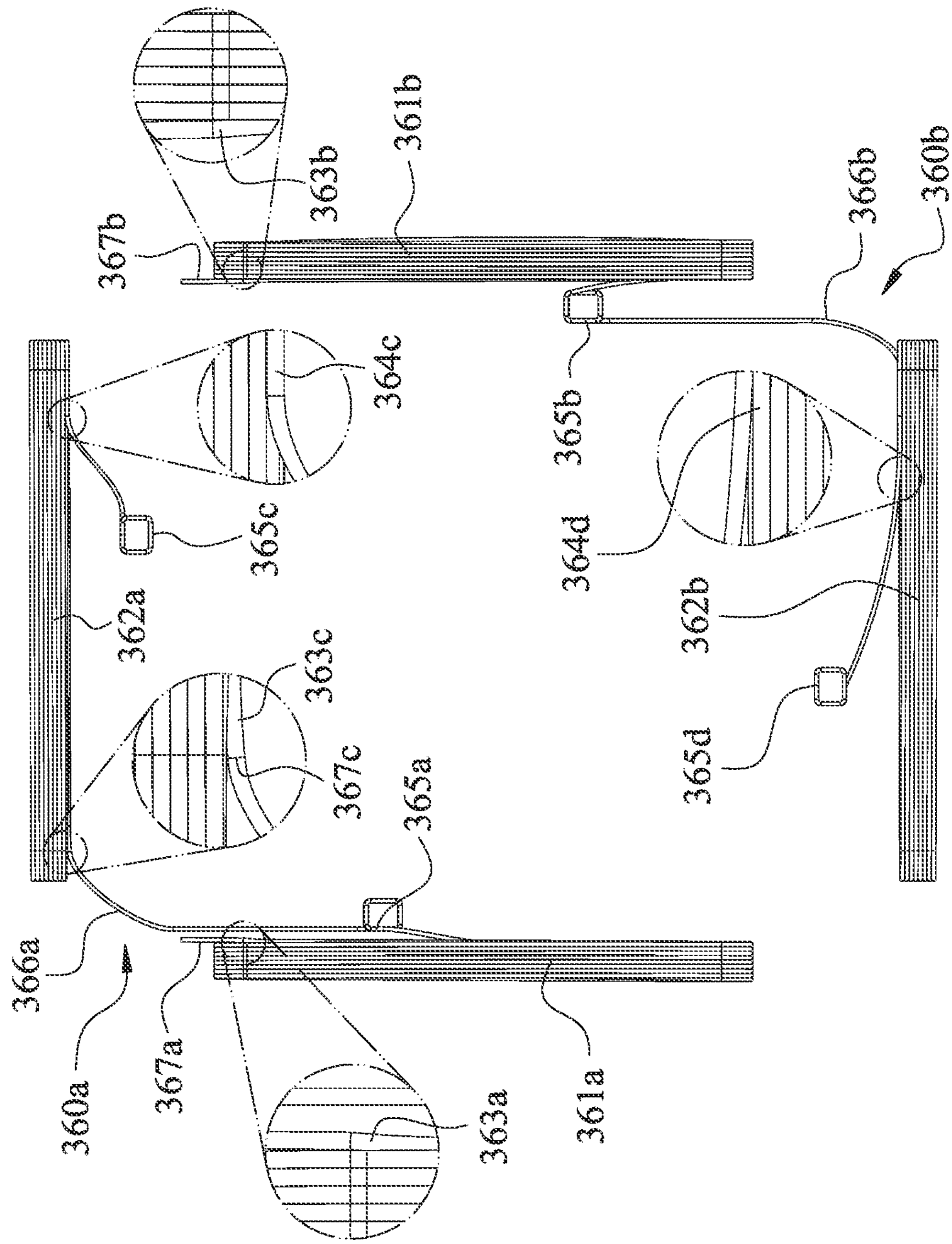
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Fig. 3E

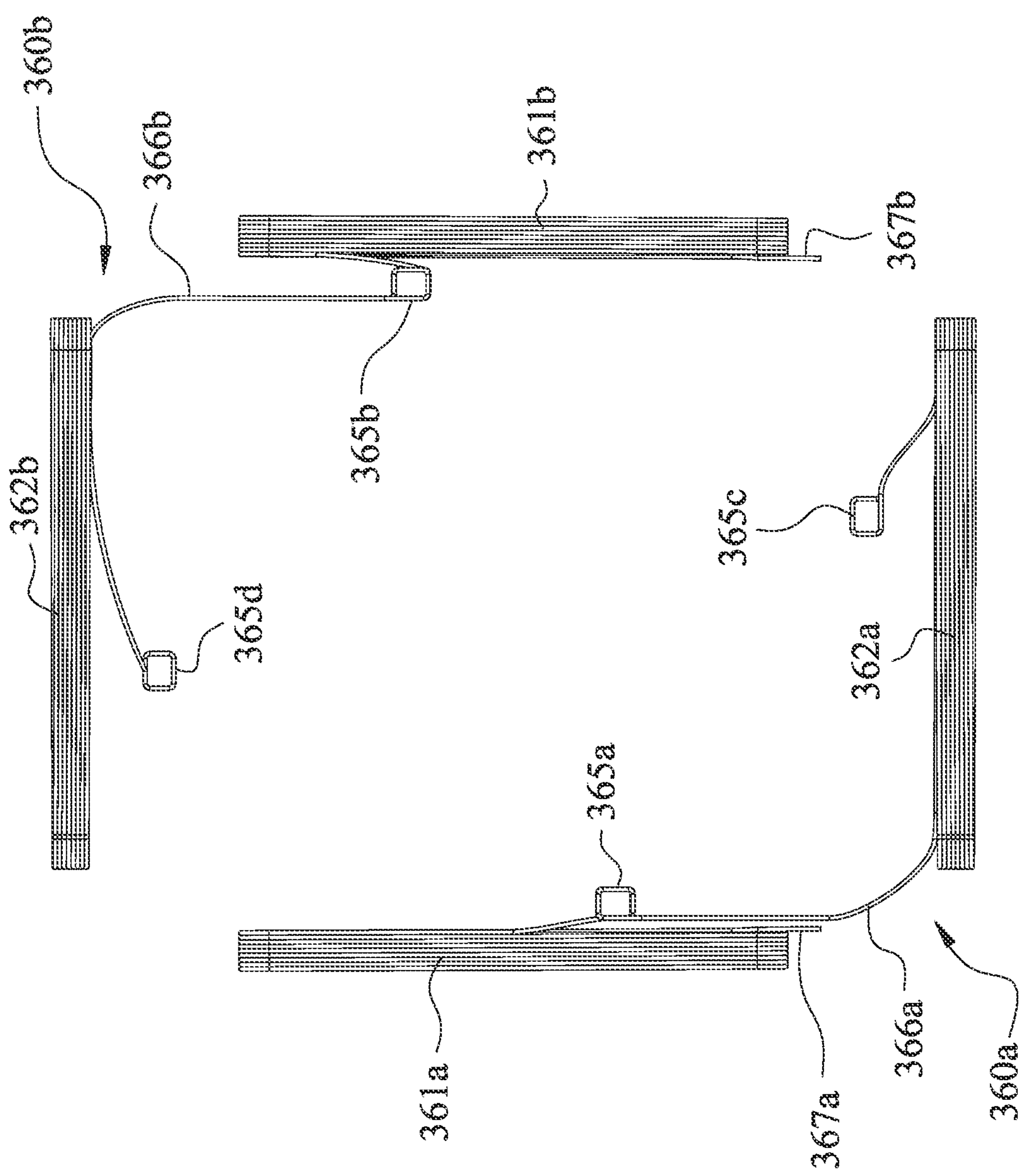

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Fig. 3F

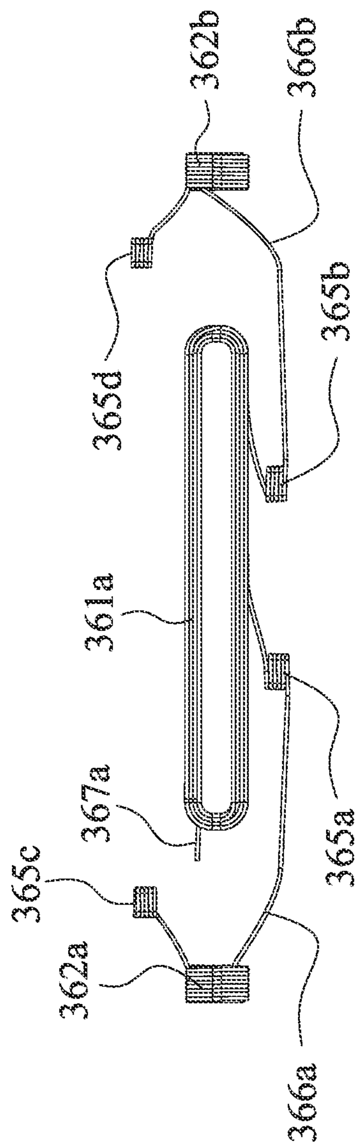
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Fig. 3G

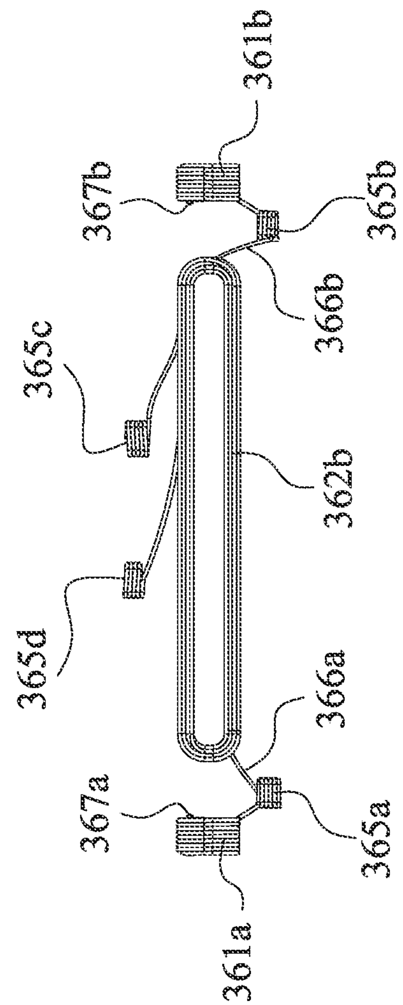
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Fig. 3H

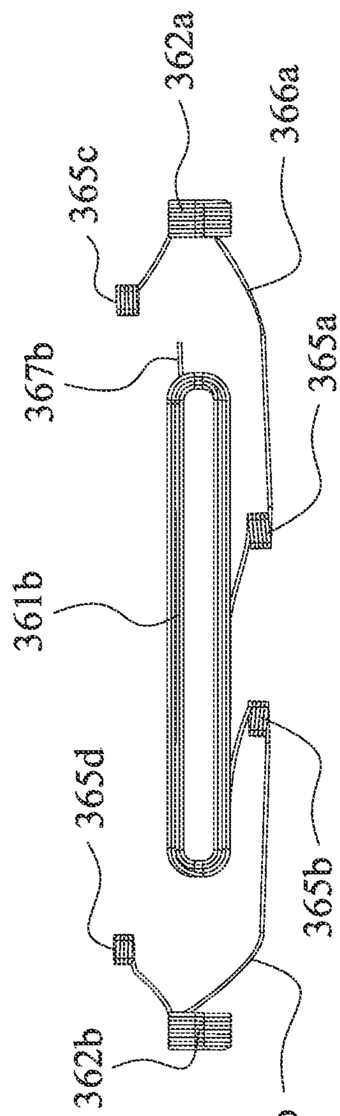

Fig. 3I

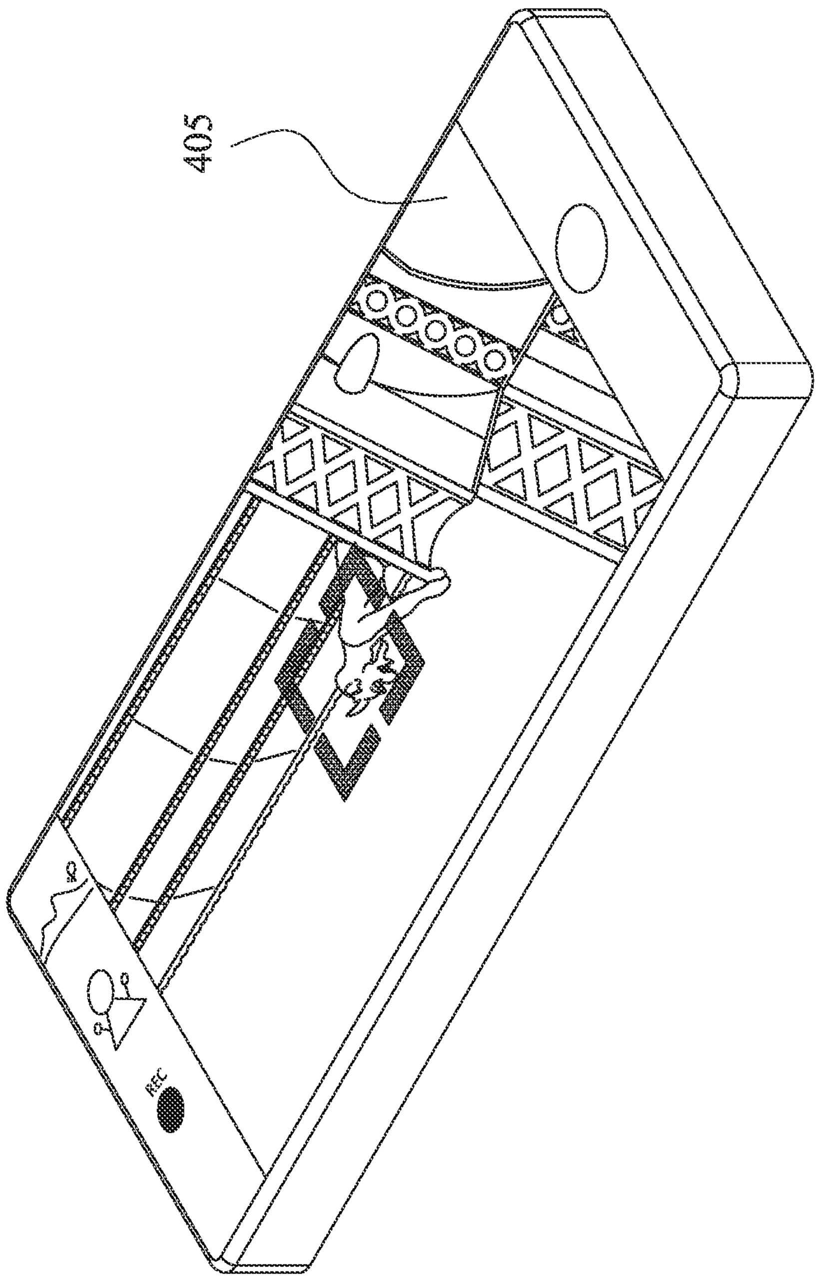

Fig. 4A

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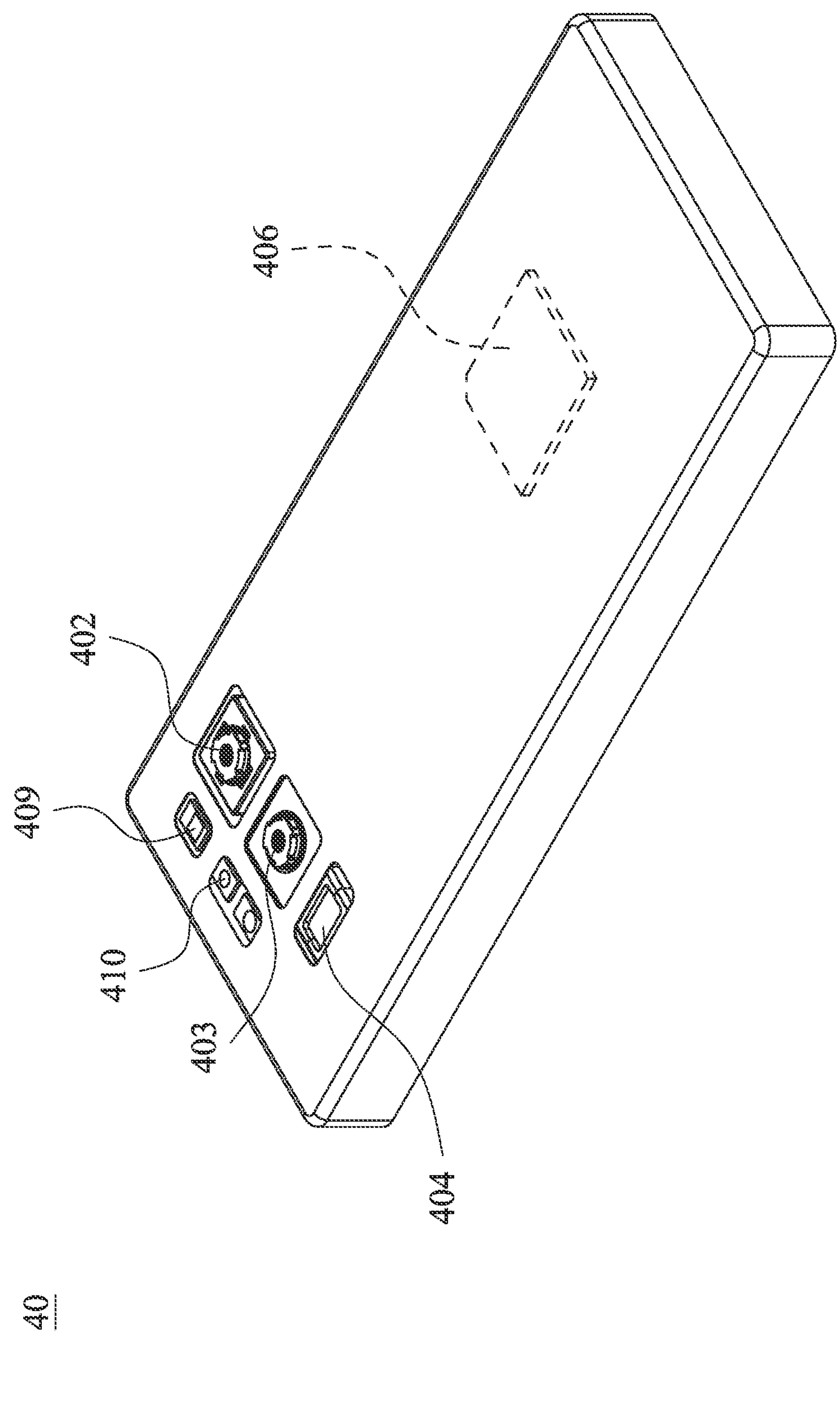
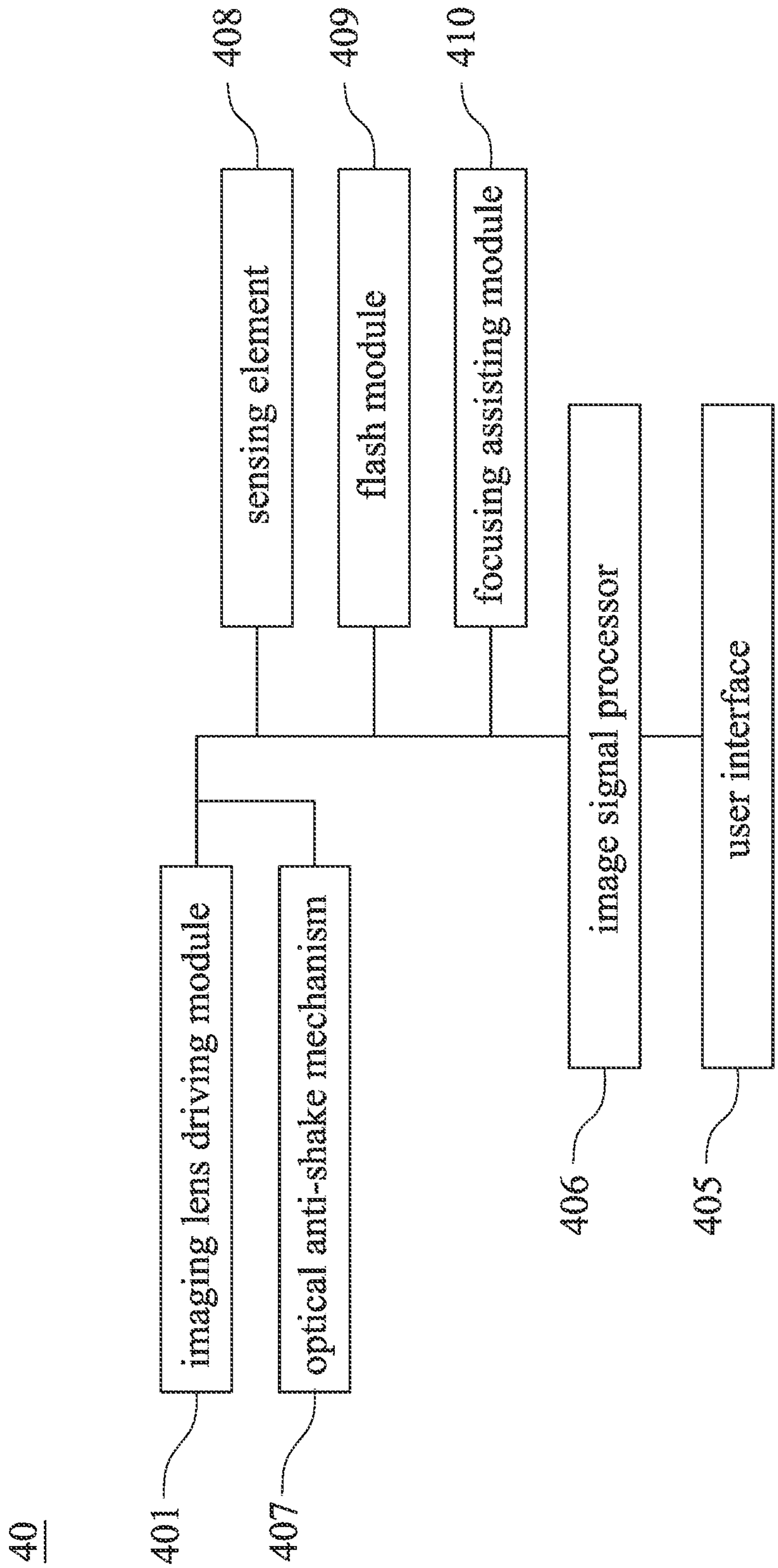


Fig. 4B


40
401
imaging lens driving module
407
optical anti-shake mechanism
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sensing element
409
flash module
410
focusing assisting module
406
image signal processor
405
user interface

Fig. 4C

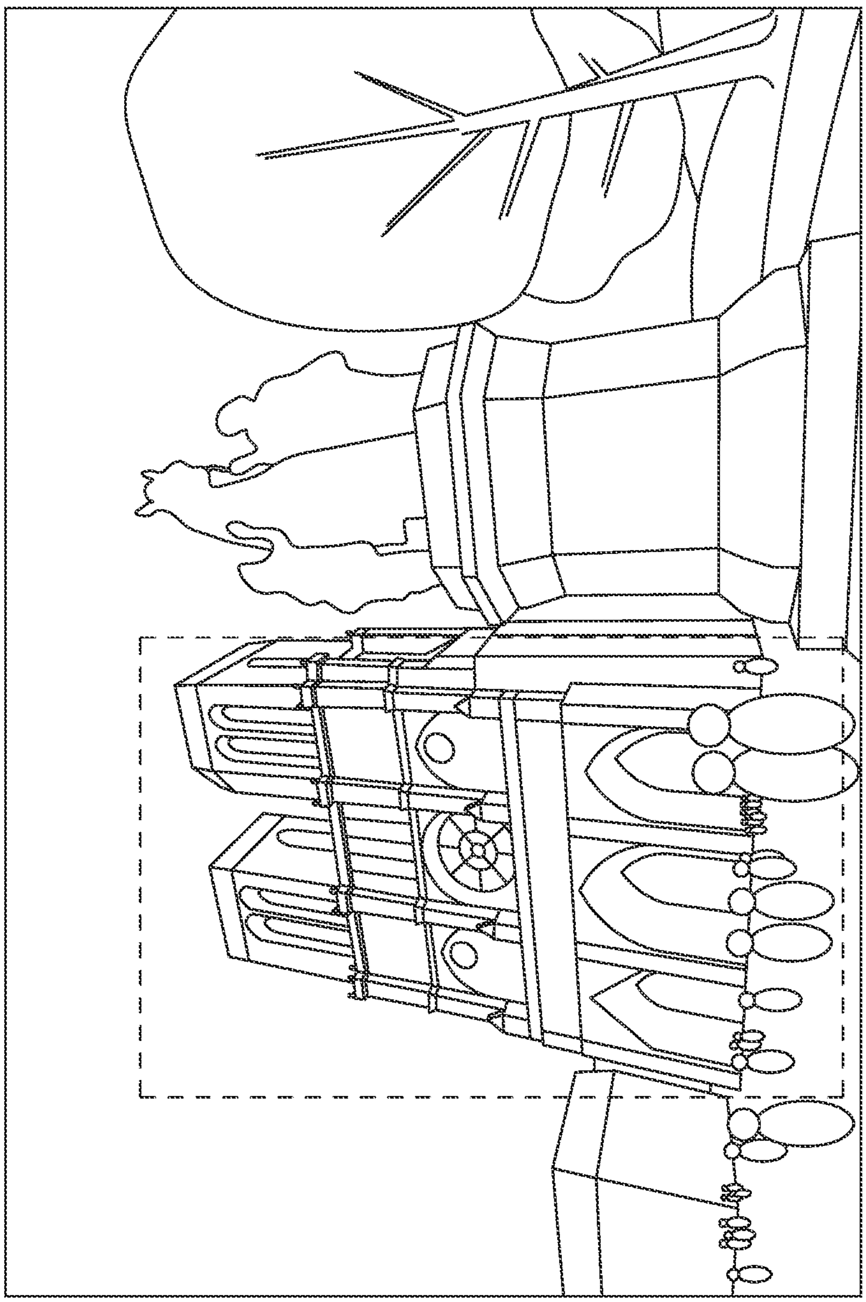

Fig. 4D

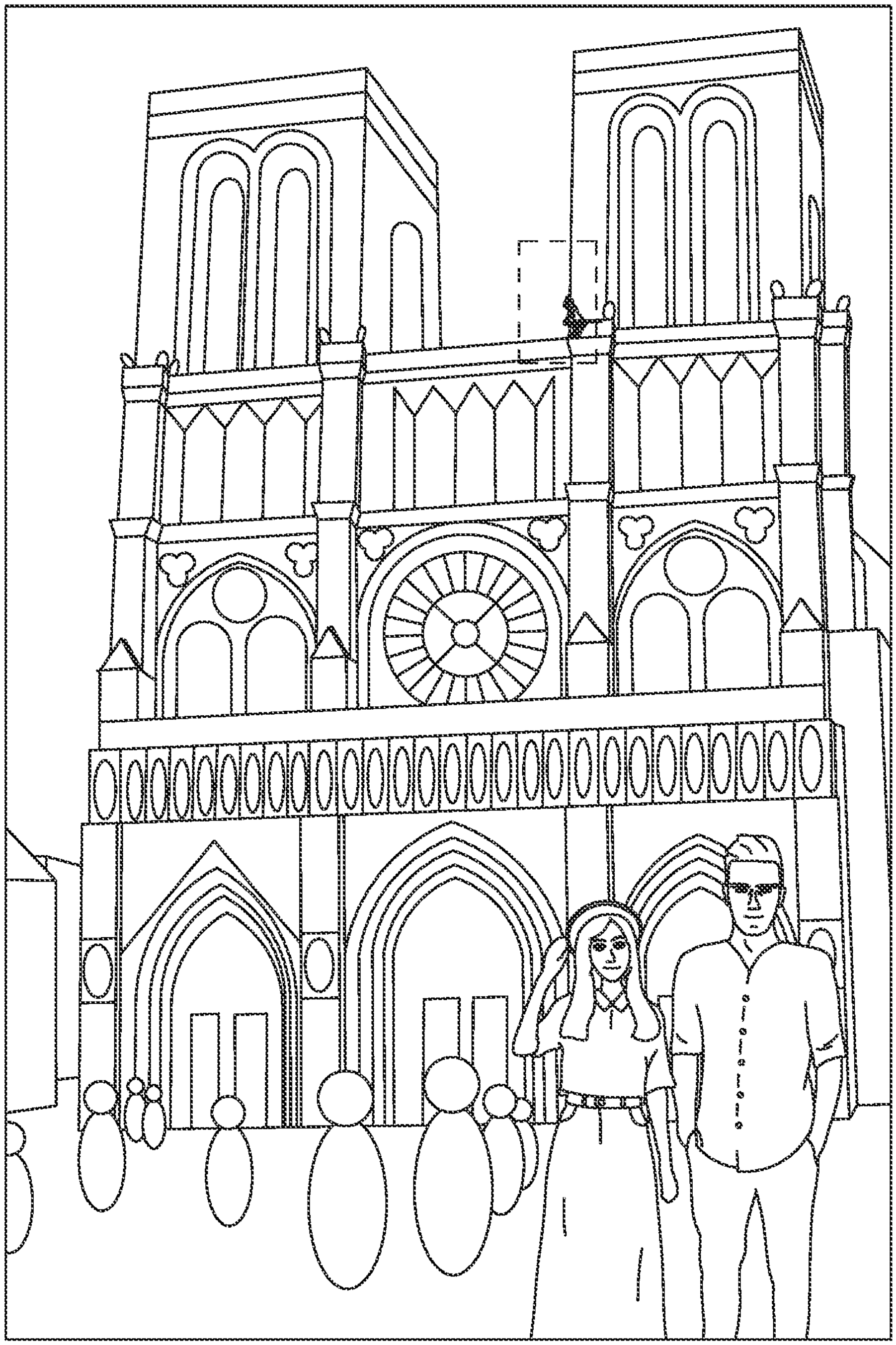

Fig. 4E

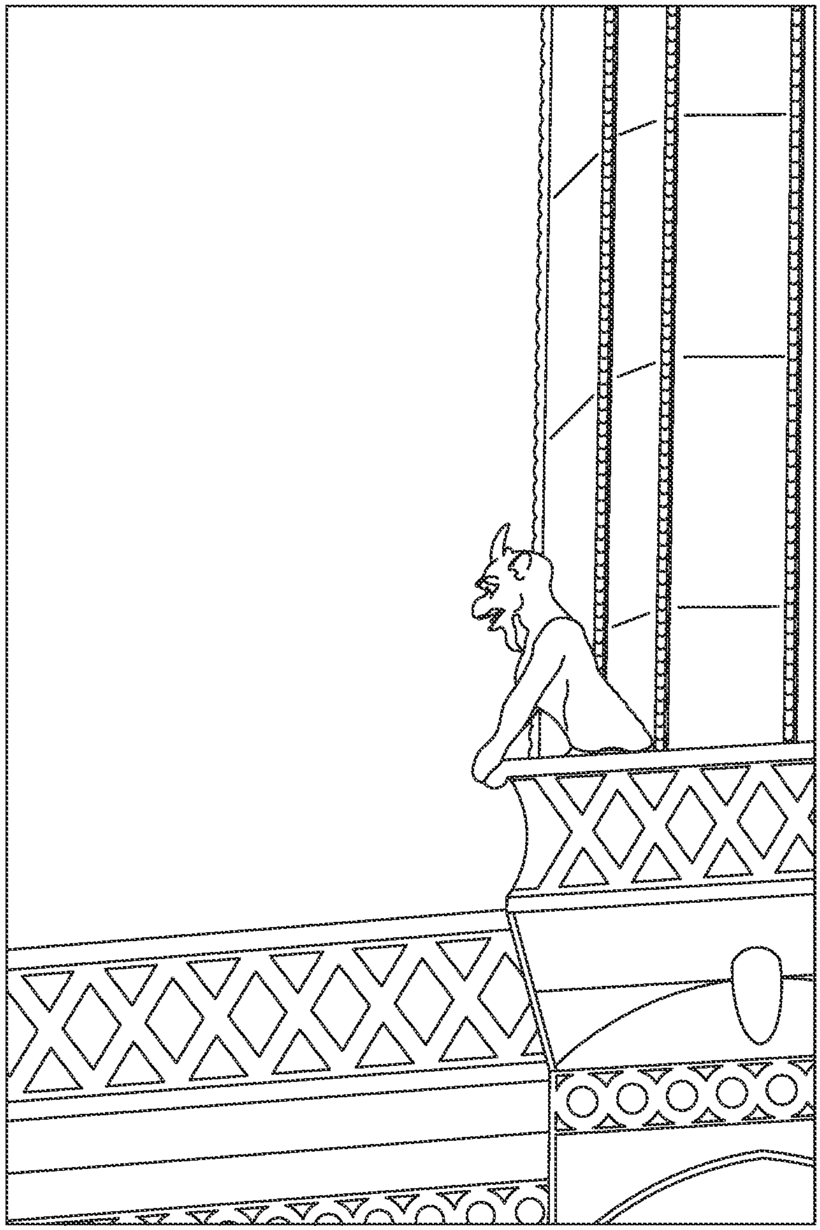

Fig. 4F

IMAGING LENS DRIVING MODULE AND ELECTRONIC DEVICE

RELATED APPLICATIONS

This application claims priority to Taiwan Application Serial Number 109137719, filed Oct. 29, 2020, which is herein incorporated by reference.

BACKGROUND

Technical Field

The present disclosure relates to an imaging lens driving module. More particularly, the present disclosure relates to an imaging lens driving module applicable to portable electronic devices.

Description of Related Art

In recent years, portable electronic devices have developed rapidly. For example, intelligent electronic devices and tablets have been filled in the lives of modern people, and imaging lens driving modules mounted on portable electronic devices have also prospered. However, as technology advances, the quality requirements of the imaging lens driving modules are becoming higher and higher. Therefore, an imaging lens driving module, which the manufacturing efficiency can be promoted, needs to be developed.

SUMMARY

According to one aspect of the present disclosure, an imaging lens driving module includes an imaging lens set, a carrier element and a driving mechanism. The imaging lens set has an optical axis. The carrier element is configured to dispose the imaging lens set, and includes an assembling structure. The assembling structure is disposed on an outer surface of the carrier element, and extends along a direction away from the optical axis. The driving mechanism is configured to drive the carrier element to move along a direction parallel to the optical axis, and includes at least one coil pair, at least two magnets and at least one elastic element. The coil pair is disposed on the assembling structure of the carrier element, and includes a bottom layer coil and a top layer coil. The bottom layer coil is wound around and directly contacted with the assembling structure. The top layer coil is stacked on and wound around the bottom layer coil, the top layer coil is farther away from the assembling structure than the bottom layer coil away from the assembling structure, and the top layer coil overlaps the bottom layer coil along the direction parallel to the optical axis. The magnets correspond to the coil pair, respectively. The elastic element is coupled with the carrier element. The coil pair only has two wire terminals, and the wire terminals are disposed on the top layer coil, respectively. The wire terminals of the top layer coil are electrically connected to the elastic element of the driving mechanism.

According to one aspect of the present disclosure, an electronic device includes the imaging lens driving module of the aforementioned aspect.

According to one aspect of the present disclosure, an imaging lens driving module includes an imaging lens set, a carrier element and a driving mechanism. The imaging lens set has an optical axis. The carrier element is configured to dispose the imaging lens set, and includes an assembling structure. The assembling structure is disposed on an outer surface of the carrier element, and extends along a direction away from the optical axis. The driving mechanism is configured to drive the carrier element to move along a direction parallel to the optical axis, and includes at least one coil pair and at least two magnets. The coil pair is disposed on the assembling structure of the carrier element, and includes a bottom layer coil and a top layer coil. The bottom layer coil is wound around and directly contacted with the assembling structure. The top layer coil is stacked on and wound around the bottom layer coil, the top layer coil is farther away from the assembling structure than the bottom layer coil away from the assembling structure, and the top layer coil overlaps the bottom layer coil along the direction parallel to the optical axis. The magnets correspond to the coil pair, respectively. The bottom layer coil of the coil pair only has two wire ends. The coil pair further includes a connecting wire connected to the wire ends of the bottom layer coil so as to keep the bottom layer coil electrically connected.

According to one aspect of the present disclosure, an imaging lens driving module includes an imaging lens set, a carrier element and a driving mechanism. The imaging lens set has an optical axis. The carrier element is configured to dispose the imaging lens set, and includes an assembling structure. The assembling structure is disposed on an outer surface of the carrier element, and extends along a direction away from the optical axis. The driving mechanism is configured to drive the carrier element to move along a direction parallel to the optical axis, and includes at least one coil pair, at least two magnets and at least one elastic element. The coil pair is disposed on the assembling structure of the carrier element, and includes a first coil and a second coil. The first coil is disposed on the assembling structure of the carrier element. The second coil is disposed on the assembling structure of the carrier element, the second coil corresponds to the first coil, and each of the first coil and the second coil includes a bottom layer coil and a top layer coil. The bottom layer coil is wound around and directly contacted with the assembling structure. The top layer coil is stacked on and wound around the bottom layer coil, the top layer coil is farther away from the assembling structure than the bottom layer coil away from the assembling structure, and the top layer coil overlaps the bottom layer coil along the direction parallel to the optical axis. The magnets correspond to the coil pair, respectively. The elastic element is coupled with the carrier element. A coiling direction of the first coil and a coiling direction of the second coil are the same during observing from the first coil towards the second coil.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A is an exploded schematic view of an imaging lens driving module according to the 1st example of the present disclosure.

FIG. **1**B is another exploded schematic view of the imaging lens driving module according to the 1st example in FIG. **1**A.

FIG. **1**C is a partial schematic view of the imaging lens driving module according to the 1st example in FIG. **1**A.

FIG. **1**D is an object-side schematic view of the carrier element and the coil pair according to the 1st example in FIG. **1**A.

FIG. **1**E is an image-side schematic view of the carrier element and the coil pair according to the 1st example in FIG. **1**A.

FIG. 1F is a side schematic view of the carrier element and the coil pair according to the 1st example in FIG. 1A.

FIG. 1G is another side schematic view of the carrier element and the coil pair according to the 1st example in FIG. 1A.

FIG. 1H is another side schematic view of the carrier element and the coil pair according to the 1st example in FIG. 1A.

FIG. 1I is another side schematic view of the carrier element and the coil pair according to the 1st example in FIG. 1A.

FIG. 1J is a partial side schematic view of the carrier element and the coil pair according to the 1st example in FIG. 1A.

FIG. 1K is a stacking and winding schematic view of the coil pair according to the 1st example in FIG. 1A.

FIG. 1L is a partial enlarged view of the carrier element and the coil pair according to the 1st example in FIG. 1A.

FIG. 1M is a schematic view of the carrier element and the coil pair according to the 1st example in FIG. 1A.

FIG. 1N is a schematic view of the coil pair according to the 1st example in FIG. 1M.

FIG. 1O is another schematic view of the coil pair according to the 1st example in FIG. 1M.

FIG. 1P is another schematic view of the carrier element and the coil pair according to the 1st example in FIG. 1A.

FIG. 1Q is a schematic view of the coil pair according to the 1st example in FIG. 1P.

FIG. 1R is another schematic view of the coil pair according to the 1st example in FIG. 1P.

FIG. 2A is a partial schematic view of an imaging lens driving module according to the 2nd example of the present disclosure.

FIG. 2B is another partial schematic view of the imaging lens driving module according to the 2nd example in FIG. 2A.

FIG. 2C is a partial side schematic view of the imaging lens driving module according to the 2nd example in FIG. 2A.

FIG. 2D is a partial object-side schematic view of the imaging lens driving module according to the 2nd example in FIG. 2A.

FIG. 2E is a partial image-side schematic view of the imaging lens driving module according to the 2nd example in FIG. 2A.

FIG. 3A is a schematic view of coil pairs according to the 3rd example of the present disclosure.

FIG. 3B is another schematic view of the coil pairs according to the 3rd example in FIG. 3A.

FIG. 3C is a schematic view of the first coils according to the 3rd example in FIG. 3A.

FIG. 3D is a schematic view of the second coils according to the 3rd example in FIG. 3A.

FIG. 3E is an object-side schematic view of the coil pairs according to the 3rd example in FIG. 3A.

FIG. 3F is an image-side schematic view of the coil pairs according to the 3rd example in FIG. 3A.

FIG. 3G is a side schematic view of the coil pairs according to the 3rd example in FIG. 3A.

FIG. 3H is another side schematic view of the coil pairs according to the 3rd example in FIG. 3A.

FIG. 3I is still another side schematic view of the coil pairs according to the 3rd example in FIG. 3A.

FIG. 4A is a schematic view of an electronic device according to the 4th example of the present disclosure.

FIG. 4B is another schematic view of the electronic device according to the 4th example in FIG. 4A.

FIG. 4C is a block diagram of the electronic device according to the 4th example in FIG. 4A.

FIG. 4D is a schematic view of an image shot via the ultra-wide angle camera module according to the 4th example in FIG. 4A.

FIG. 4E is a schematic view of an image shot via the high resolution camera module according to the 4th example in FIG. 4A.

FIG. 4F is a schematic view of an image shot via the telephoto camera module according to the 4th example in FIG. 4A.

DETAILED DESCRIPTION

The present disclosure provides an imaging lens driving module, and the imaging lens driving module includes an imaging lens set, a carrier element and a driving mechanism. The imaging lens set has an optical axis. The carrier element is configured to dispose the imaging lens set, and includes an assembling structure, wherein the assembling structure is disposed on an outer surface of the carrier element, and extends along a direction away from the optical axis. The driving mechanism is configured to drive the carrier element to move along a direction parallel to the optical axis, and includes at least one coil pair and at least two magnets. The coil pair is disposed on the assembling structure of the carrier element, and includes a bottom layer coil and a top layer coil, wherein the bottom layer coil is wound around and directly contacted with the assembling structure; the top layer coil is stacked on and wound around the bottom layer coil, the top layer coil is farther away from the assembling structure than the bottom layer coil away from the assembling structure, and the top layer coil overlaps the bottom layer coil along the direction parallel to the optical axis. The magnets correspond to the coil pair, respectively. In particular, according to the present disclosure, the same wire is wound around the assembling structure along the same coiling direction, and the coil pair, which is relatively disposed, is formed. In the conventional art, a coil pair is not formed by simultaneously winding around an assembling structure. Hence, the cycle time of the product can be reduced by the imaging lens driving module of the present disclosure relative to the coil pair of the conventional art.

In detail, the carrier element can be a coil holder, a lens carrier or elements which are combined with the aforementioned functions, but the present disclosure is not limited thereto. Further, a driving magnetic force can be generated by the interaction between the magnets and the coil pair, and the carrier element can be driven to move along the direction parallel to the optical axis by the driving mechanism.

The driving mechanism can further include at least one elastic element coupled with the carrier element. Furthermore, a number of the elastic element can be two, and the elastic elements can include an upper elastic element and a lower elastic element. The upper elastic element is disposed on an object side of the imaging lens set, the lower elastic element is disposed on an image side of the imaging lens set, and the lower elastic element corresponds to the upper elastic element. Therefore, the driving range of the driving mechanism can be defined.

The coil pair can only have two wire terminals, and the wire terminals are disposed on the top layer coil, respectively. In particular, the wire terminals can be a wire wound around an end area of the carrier element, and the wire terminals can be disconnected. Further, an end of the wire terminals is connected to the top layer coil, and another end of the wire terminals is exposed in the air.

The wire terminals of the top layer coil can be electrically connected to the elastic element of the driving mechanism, wherein the wire terminals of the top layer coil can be electrically connected to the elastic element of the driving mechanism by welding or dispensing, but the present disclosure is not limited thereto. Therefore, the wire terminals can be electrically connected to other elements so as to promote the design margin of the product station. In particular, the performing sequence can be adjusted relatively to other stations while the planning and design stage of the welding station or the dispensing station.

The coil pair of the driving mechanism can be the wire composed by simultaneously winding around the assembling structure of the carrier element from the bottom layer coil towards the wire terminals of the top layer coil. In detail, the bottom layer coil is composed of the wire, and the top layer coil has the wire terminals, hence, the coil pair is composed of the same wire. Forming the coil pair by the same wire simultaneously winding around is favorable for reducing the manufacturing process and lowering the cycle time of the product.

The carrier element can further include at least two columnar structures, wherein the columnar structures are disposed on the outer surface of the carrier element, and extend along the direction parallel to the optical axis and a direction towards an image side of the imaging lens set. In particular, the columnar structures and the carrier element can be integrally formed, but the present disclosure is not limited thereto. The assembling tolerance between the elements can be reduced by the integral formation of the columnar structures and the carrier element.

Each of the wire terminals of the top layer coil can be wound around and directly contacted with each of the columnar structures. In detail, the required tension of the wire for fixing the coil pair is provided by the wire terminals wound around the columnar structures. Therefore, the wire can be prevented from loosing from the carrier element to promote the yield rate.

The lower elastic element can include an inner-side portion, an outer-side portion and an elastic connecting portion, wherein the inner-side portion is coupled with the carrier element, the outer-side portion is coupled with a base of the imaging lens driving module, and the elastic connecting portion is connected to the inner-side portion and the outer-side portion. Therefore, the lower elastic element can have the better coupled location to ensure the driving efficiency of the driving mechanism.

The wire terminals of the top layer coil can be electrically connected to the lower elastic element except the outer-side portion. Therefore, the better location of the electrical connection can be obtained, and the aforementioned disposition is favorable for the compact size of the imaging lens driving module.

The carrier element can further include at least one abutting portion, wherein the abutting portion and the assembling structure are alternately disposed along a circumferential direction around the optical axis. Therefore, the space utilization in the imaging lens driving module can be enhanced.

The connecting wire of the coil pair can be directly contacted with a separation point of the abutting portion of the carrier element. By the separation point, the process of the automatic optical inspection can be more precise and faster, and the wire length of the coil pair can be effectively controlled to lower the manufacturing cost.

Or, the bottom layer coil of the coil pair can only have two wire ends, wherein the coil pair can further include a connecting wire connected to the wire ends of the bottom layer coil so as to keep the bottom layer coil electrically connected. In detail, the connecting wire can be a portion of the coil pair, but the present disclosure is not limited thereto. Therefore, the feasibility of the coil pair composed of the wire which is continuous can be provided.

The coil pair can further include a first coil and a second coil, wherein the first coil is disposed on the assembling structure of the carrier element, the second coil is disposed on the assembling structure of the carrier element, and the second coil corresponds to the first coil. Moreover, each of the first coil and the second coil can include the bottom layer coil and the top layer coil, and a coiling direction of the first coil and a coiling direction of the second coil are the same during observing from the first coil towards the second coil, that is, both of the coiling direction of the first coil and the coiling direction of the second coil are clockwise or counterclockwise. By the same coiling direction, the coiling process can be simplified to further reduce the time cost.

The first coil and the second coil can be composed of two wires, so as to keep the first coil and the second coil electrically separated. In detail, the electrical separation between the first coil and the second coil can be generated by composing the first coil and the second coil, which are different wires simultaneously wound around the assembling structure, and the tilt of the imaging lens set can be adjusted. Therefore, the first coil and the second coil, which are electrically separated, can be separately controlled to further improve the imaging quality.

The lower elastic element can include four elastic sheets electrically separated from each other. In particular, the elastic sheets are suitable for the mass production and the assembling, and the volume of the imaging lens driving module can be effectively reduced via the elastic sheets. Further, the elastic sheets can be electrically connected to the wire terminal of the first coil, the wire end of the first coil, the wire terminal of the second coil and the wire end of the second coil, respectively. In detail, the elastic sheets can be further electrically connected to the wire terminal of the top layer coil of the first coil, the wire end of the bottom layer coil of the first coil, the wire terminal of the top layer coil of the second coil and the wire end of the bottom layer coil of the second coil, respectively. Therefore, the elastic sheets can be kept in electrical separation, wherein two of the elastic sheets can be electrically connected to the first coil, and the other two of the elastic sheets can be electrically connected to the second coil. Hence, the precision of the tilt of the imaging lens set can be enhanced.

The coil pair includes a number of coil pairs and a number of wire terminals, the number of the coil pairs is M, the number of the wire terminals is N, and the following conditions can be satisfied: $N/2=M$; and $2\leq N\leq 10$. Therefore, the suitable range of the number of the coil pair can be obtained.

Each of the aforementioned features of the imaging lens driving module can be utilized in various combinations for achieving the corresponding effects.

The present disclosure provides an electronic device, which includes the aforementioned imaging lens driving module.

According to the aforementioned embodiment, specific examples are provided, and illustrated via figures.

1st Example

FIG. **1A** is an exploded schematic view of an imaging lens driving module **100** according to the 1st example of the present disclosure. FIG. **1**B is another exploded schematic view of the imaging lens driving module **100** according to the 1st example in FIG. **1**A. In FIGS. **1**A and **1**B, the imaging lens driving module **100** includes a cover **110**, a gasket **120**, a driving mechanism (its reference is omitted), an imaging lens set **140**, a carrier element **150** and a base **190**, wherein the imaging lens set **140** has an optical axis X.

The carrier element **150** is configured to dispose the imaging lens set **140**, and includes an assembling structure **151**, at least two columnar structures **152** and at least one abutting portion **153**. The assembling structure **151** is disposed on an outer surface of the carrier element **150**, and extends along a direction away from the optical axis X. The columnar structures **152** are disposed on the outer surface of the carrier element **150**, and extend along a direction parallel to the optical axis X and a direction towards an image side of the imaging lens set **140**. The abutting portion **153** and the assembling structure **151** are alternately disposed along a circumferential direction around the optical axis X. In detail, the carrier element **150** can be a coil holder, a lens carrier or elements which are combined with the aforementioned functions, and the columnar structures **152** and the carrier element **150** can be integrally formed, but the present disclosure is not limited thereto. The assembling tolerance between the elements can be reduced by the integral formation of the columnar structures **152** and the carrier element **150**. Furthermore, the space utilization in the imaging lens driving module **100** can be enhanced by the alternate disposition of the abutting portion **153** and the assembling structure **151** along the circumferential direction around the optical axis X.

The driving mechanism is configured to drive the carrier element **150** to move along the direction parallel to the optical axis X, and includes at least one elastic element, at least one coil pair **160** and at least two magnets **170**. The elastic element is coupled with the carrier element **150**. A number of the elastic element can be two, and the elastic elements include an upper elastic element **130** and a lower elastic element **180**. The upper elastic element **130** is disposed on an object side of the imaging lens set **140**, the lower elastic element **180** is disposed on the image side of the imaging lens set **140**, and the lower elastic element **180** corresponds to the upper elastic element **130**. Therefore, the driving range of the driving mechanism can be defined. The coil pair **160** is disposed on the assembling structure **151** of the carrier element **150**. The magnets **170** correspond to the coil pair **160**, respectively. A driving magnetic force can be generated by the interaction between each of the magnets **170** and the coil pair **160**, and the carrier element **150** can be driven to move along the direction parallel to the optical axis X by the driving mechanism. It should be mentioned that the gasket **120** can be configured to adjust the location of the upper elastic element **130** relative to the cover **110**.

The lower elastic element **180** includes an inner-side portion **181**, an outer-side portion **182** and an elastic connecting portion **183**, wherein the inner-side portion **181** is coupled with the carrier element **150**, the outer-side portion **182** is coupled with the base **190** of the imaging lens driving module **100**, and the elastic connecting portion **183** is connected to the inner-side portion **181** and the outer-side portion **182**. Therefore, the lower elastic element **180** can have the better coupled location to ensure the driving efficiency of the driving mechanism.

FIG. **1**C is a partial schematic view of the imaging lens driving module **100** according to the 1st example in FIG. **1**A. FIG. **1**D is an object-side schematic view of the carrier element **150** and the coil pair **160** according to the 1st example in FIG. **1**A. FIG. **1**E is an image-side schematic view of the carrier element **150** and the coil pair **160** according to the 1st example in FIG. **1**A. FIGS. **1**F to **1**I are side schematic views of the carrier element **150** and the coil pair **160** according to the 1st example in FIG. **1**A. FIG. **1**J is a partial side schematic view of the carrier element **150** and the coil pair **160** according to the 1st example in FIG. **1**A. FIG. **1**K is a stacking and winding schematic view of the coil pair **160** according to the 1st example in FIG. **1**A. FIG. **1**L is a partial enlarged view of the carrier element **150** and the coil pair **160** according to the 1st example in FIG. **1**A. In FIGS. **1**C to **1**L, the coil pair **160** includes a bottom layer coil **163** and a top layer coil **164**. The bottom layer coil **163** is wound around and directly contacted with the assembling structure **151**. The top layer coil **164** is stacked on and wound around the bottom layer coil **163** in a stacking and winding direction D**2**. The top layer coil **164** is farther away from the assembling structure **151** than the bottom layer coil **163** away from the assembling structure **151**, and the top layer coil **164** overlaps the bottom layer coil **163** along the direction parallel to the optical axis X.

The coil pair **160** only has two wire terminals **165**, and the wire terminals **165** are disposed on the top layer coil **164**, respectively. The wire terminals **165** of the top layer coil **164** are electrically connected to the elastic element of the driving mechanism, wherein the wire terminals **165** of the top layer coil **164** can be electrically connected to the elastic element of the driving mechanism by welding or dispensing, but the present disclosure is not limited thereto. In FIG. **1**C, according to the 1st example, the wire terminals **165** of the top layer coil **164** are electrically connected to the lower elastic element **180** of the driving mechanism, and the wire terminals **165** of the top layer coil **164** can be electrically connected to the lower elastic element **180** of the driving mechanism by dispensing to form an electrical connecting portion C connected to the lower elastic element **180**. Therefore, the wire terminals **165** can be electrically connected to other elements so as to promote the design margin of the product station.

Furthermore, the wire terminals **165** of the top layer coil **164** are electrically connected to the lower elastic element **180** except the outer-side portion **182**. That is, the wire terminals **165** of the top layer coil **164** are electrically connected to the inner-side portion **181** of the lower elastic element **180** and the elastic connecting portion **183** of the lower elastic element **180**, but the present disclosure is not limited thereto. Therefore, the better location of the electrical connection can be obtained, and the aforementioned disposition is favorable for the compact size of the imaging lens driving module **100**.

In FIGS. **1**E, **1**F and **1**L, the coil pair **160** of the driving mechanism can be a wire composed by simultaneously winding around the assembling structure **151** of the carrier element **150** from the bottom layer coil **163** towards the wire terminals **165** of the top layer coil **164** along a coiling direction D**1**. In detail, the bottom layer coil **163** is composed by the wire, and the top layer coil **164** has the wire terminals **165**. Hence, the coil pair **160** is composed of the same wire. Forming the coil pair **160** by the same wire simultaneously winding around is favorable for reducing the manufacturing process and lowering the cycle time of the product.

In FIGS. **1**C and **1**E, each of the wire terminals **165** of the top layer coil **164** can be wound around and directly contacted with each of the columnar structures **152**. In particular, the wire terminals **165** can be the wire wound around an end area (that is, the columnar structure **152**) of the carrier element **150**, and the wire terminals **165** can be disconnected, and one of the wire terminals **165** is exposed in the air. In detail, the required tension of the wire for fixing the coil pair **160** is provided by the wire terminals **165** of the top layer coil **164** wound around the columnar structure **152**. Therefore, the wire can be prevented from loosing from the carrier element **150** to promote the yield rate.

FIG. **1**M is a schematic view of the carrier element **150** and the coil pair **160** according to the 1st example in FIG. **1**A. FIGS. **1**N and **1**O are schematic views of the coil pair **160** according to the 1st example in FIG. **1**M. FIG. **1**P is another schematic view of the carrier element **150** and the coil pair **160** according to the 1st example in FIG. **1**A. FIGS. **1**Q and **1**R are schematic views of the coil pair **160** according to the 1st example in FIG. **1**P. In FIGS. **1**L to **1**R, the bottom layer coil **163** of the coil pair **160** only has two wire ends **16*x*1**, and the coil pair **160** further includes a connecting wire **166**, wherein each of two wire ends **1631** of the connecting wire **166** is connected to each of the wire ends **16*x*1** of the bottom layer coil **163** so as to keep the bottom layer coil **163** electrically connected.

Moreover, the connecting wire **166** can be a portion of the coil pair **160**, but the present disclosure is not limited thereto. Therefore, the feasibility of the coil pair **160** composed of the wire which is continuous can be provided.

The connecting wire **166** of the coil pair **160** is directly contacted with a separation point (its reference numeral is omitted) of the abutting portion **153** of the carrier element **150**. By the separation point, the process of the automatic optical inspection can be more precise and faster, and the wire length of the coil pair **160** can be effectively controlled to lower the manufacturing cost.

According to the 1st example, the coil pair includes one coil pair and two wire terminals, but the present disclosure is not limited thereto.

2nd Example

FIG. **2**A is a partial schematic view of an imaging lens driving module according to the 2nd example of the present disclosure. FIG. **2**B is another partial schematic view of the imaging lens driving module according to the 2nd example in FIG. **2**A. In FIGS. **2**A and **2**B, the imaging lens driving module (its reference is omitted) includes a cover (not shown), a gasket (not shown), a driving mechanism (its reference is omitted), an imaging lens set (not shown), a carrier element **250** and a base (not shown), wherein the imaging lens set has an optical axis (its reference is omitted).

The carrier element **250** is configured to dispose the imaging lens set, and includes an assembling structure **251** (as shown in FIG. **2**D), at least two columnar structures **252** and at least one abutting portion **253** (as shown in FIG. **2**D). The assembling structure **251** is disposed on an outer surface of the carrier element **250**, and extends along a direction away from the optical axis. The columnar structures **252** are disposed on the outer surface of the carrier element **250**, and extend along a direction parallel to the optical axis and a direction towards an image side of the imaging lens set. The abutting portion **253** and the assembling structure **251** are alternately disposed along a circumferential direction around the optical axis. In detail, the carrier element **250** can be a coil holder, a lens carrier or elements which are combined with the aforementioned functions, and the columnar structures **252** and the carrier element **250** can be integrally formed, but the present disclosure is not limited thereto. The assembling tolerance between the elements can be reduced by the integral formation of the columnar structures **252** and the carrier element **250**. Furthermore, the space utilization in the imaging lens driving module can be enhanced by the alternate disposition of the abutting portion **253** and the assembling structure **251** along the circumferential direction around the optical axis.

The driving mechanism is configured to drive the carrier element **250** to move along the direction parallel to the optical axis, and includes at least one elastic element, at least one coil pair (its reference numeral is omitted) and at least two magnets (not shown). The elastic element is coupled with the carrier element **250**. A number of the elastic element can be two, and the elastic elements include an upper elastic element **230** and a lower elastic element **280**. The upper elastic element **230** is disposed on an object side of the imaging lens set, the lower elastic element **280** is disposed on the image side of the imaging lens set, and the lower elastic element **280** corresponds to the upper elastic element **230**. Therefore, the driving range of the driving mechanism can be defined. The coil pair is disposed on the assembling structure **251** of the carrier element **250**.

The lower elastic element **280** can include four elastic sheets (its reference numeral is omitted) electrically separated from each other. In particular, the elastic sheets are suitable for the mass production and the assembling, and the volume of the imaging lens driving module can be effectively reduced via the elastic sheets.

FIG. **2**C is a partial side schematic view of the imaging lens driving module according to the 2nd example in FIG. **2**A. FIG. **2**D is a partial object-side schematic view of the imaging lens driving module according to the 2nd example in FIG. **2**A. FIG. **2**E is a partial image-side schematic view of the imaging lens driving module according to the 2nd example in FIG. **2**A. In FIGS. **2**A to **2**E, the coil pair includes a first coil **261** and a second coil **262**, wherein the first coil **261** is disposed on the assembling structure **251** of the carrier element **250**, the second coil **262** is disposed on the assembling structure **251** of the carrier element **250**, and the second coil **262** corresponds to the first coil **261**.

Each of the first coil **261** and the second coil **262** includes a bottom layer coil (its reference numeral is omitted) and a top layer coil (its reference numeral is omitted), wherein the bottom layer coil is wound around and directly contacted with the assembling structure **251**; the top layer coil is stacked on and wound around the bottom layer coil, the top layer coil is farther away from the assembling structure **251** than the bottom layer coil away from the assembling structure **251**, and the top layer coil overlaps the bottom layer coil along the direction parallel to the optical axis.

In FIG. **2**E, a coiling direction D1 of the first coil **261** and a coiling direction D1 of the second coil **262** are the same during observing from the first coil **261** towards the second coil **262**. In particular, both of the coiling direction D1 of the first coil and the coiling direction D1 of the second coil are clockwise or counterclockwise. By the same coiling direction D1, the coiling process can be simplified to further reduce the time cost.

The first coil **261** and the second coil **262** can be composed of two wires, so as to keep the first coil **261** and the second coil **262** electrically separated. In detail, the electrical separation between the first coil **261** and the second coil **262** can be generated by composing the first coil **261** and the second coil **262**, which are different wires simultaneously wound around the assembling structure **251**, and the tilt of the imaging lens set can be adjusted. Therefore, the first coil **261** and the second coil **262**, which are electrically separated, can be separately controlled to further improve the imaging quality.

The elastic sheets can be electrically connected to a wire terminal **2611** of the first coil **261** and a wire terminal **2621** of the second coil **262**, respectively. Elastic sheet connecting portions **285** of the elastic sheets can be electrically connected to a wire end (its reference numeral is omitted) of the first coil **261** and a wire end (its reference numeral is omitted) of the second coil **262**, respectively. Further, the electrical connection is welding or dispensing, but the present disclosure is not limited thereto. Moreover, the elastic sheets can be further electrically connected to the wire terminal **2611** of the top layer coil of the first coil **261**, the wire end of the bottom layer coil of the first coil **261**, the wire terminal **2621** of the top layer coil of the second coil **262** and the wire end of the bottom layer coil of the second coil **262**, respectively. According to the 2nd example, the elastic sheets are electrically connected to the wire terminal **2611** of the first coil **261** and the wire terminal **2621** of the second coil **262** by welding. The elastic sheet connecting portions **285** of the elastic sheets are electrically connected to the wire end of the first coil **261** and the wire end of the second coil **262** by welding. Therefore, the elastic sheets can be kept in electrical separation, wherein two of the elastic sheets can be electrically connected to the first coil **261**, and the other two of the elastic sheets can be electrically connected to the second coil **262**. Hence, the precision of the tilt of the imaging lens set can be enhanced.

Furthermore, the required tension of the wire for fixing the coil pair is provided by the wire terminal **2611** of the top layer coil of the first coil **261** and the wire terminal **2621** of the top layer coil of the second coil **262** wound around the columnar structures **252** of the carrier element **250**. Therefore, the wire can be prevented from loosing from the carrier element **250** to promote the yield rate.

According to the 2nd example, the coil pair includes one coil pair and two wire terminals, but the present disclosure is not limited thereto.

Further, all of other structures and dispositions according to the 2nd example are the same as the structures and the dispositions according to the 1st example, and will not be described again herein.

3rd Example

FIG. **3**A is a schematic view of coil pairs **360***a*, **360***b* according to the 3rd example of the present disclosure. FIG. **3**B is another schematic view of the coil pairs **360***a*, **360***b* according to the 3rd example in FIG. **3**A. In FIGS. **3**A and **3**B, the coil pair **360***a* includes a first coil **361***a* and a second coil **362***a*, and the coil pair **360***b* includes a first coil **361***b* and a second coil **362***b*, wherein the first coils **361***a*, **361***b* are disposed on a carrier element (not shown) of a driving mechanism (not shown), the second coils **362***a*, **362***b* are disposed on the carrier element of the driving mechanism, and the second coils **362***a*, **362***b* and the first coils **361***a*, **361***b* are alternately disposed, respectively.

Each of the first coil and the second coil includes a bottom layer coil and a top layer coil. According to the 3rd example, the first coil **361***a* includes a bottom layer coil **363***a* and a top layer coil **364***a*, and the second coil **362***a* includes a bottom layer coil **363***c* and a top layer coil **364***c*; the first coil **361***b* includes a bottom layer coil **363***b* (as shown in FIG. **3**E) and a top layer coil **364***b*, and the second coil **362***b* includes a bottom layer coil (its reference numeral is omitted) and a top layer coil **364***d*.

Furthermore, the bottom layer coil **363***a* of the first coil **361***a*, the bottom layer coil **363***c* of the second coil **362***a*, the bottom layer coil **363***b* of the first coil **361***b* and the bottom layer coil of the second coil **362***b* are wound around and directly contacted with the assembling structure (not shown).

The top layer coil **364***a* of the first coil **361***a* is stacked on and wound around the bottom layer coil **363***a* of the first coil **361***a*, the top layer coil **364***c* of the second coil **362***a* is stacked on and wound around the bottom layer coil **363***c* of the second coil **362***a*, the top layer coil **364***b* of the first coil **361***b* is stacked on and wound around the bottom layer coil **363***b* of the first coil **361***b*, the top layer coil **364***d* of the second coil **362***b* is stacked on and wound around the bottom layer coil of the second coil **362***b*. Hence, the top layer coil **364***a* of the first coil **361***a* is farther away from the assembling structure than the bottom layer coil **363***a* of the first coil **361***a* away from the assembling structure, and the top layer coil **364***a* of the first coil **361***a* overlaps the bottom layer coil **363***a* of the first coil **361***a* along the direction parallel to an optical axis (its reference numeral is omitted); the top layer coil **364***c* of the second coil **362***a* is farther away from the assembling structure than the bottom layer coil **363***c* of the second coil **362***a* away from the assembling structure, and the top layer coil **364***c* of the second coil **362***a* overlaps the bottom layer coil **363***c* of the second coil **362***a* along the direction parallel to the optical axis; the top layer coil **364***b* of the first coil **361***b* is farther away from the assembling structure than the bottom layer coil **363***b* of the first coil **361***b* away from the assembling structure, and the top layer coil **364***b* of the first coil **361***b* overlaps the bottom layer coil **363***b* of the first coil **361***b* along the direction parallel to the optical axis; the top layer coil **364***d* of the second coil **362***b* is farther away from the assembling structure than the bottom layer coil of the second coil **362***b* away from the assembling structure, and the top layer coil **364***d* of the second coil **362***b* overlaps the bottom layer coil of the second coil **362***b* along the direction parallel to the optical axis.

FIG. **3**C is a schematic view of the first coils **361***a*, **361***b* according to the 3rd example in FIG. **3**A. FIG. **3**D is a schematic view of the second coils **362***a*, **362***b* according to the 3rd example in FIG. **3**A. In FIGS. **3**A to **3**D, the first coil **361***a* of the coil pair **360***a* and the first coil **361***b* of the coil pair **360***b* are simultaneously wound around, and then the second coil **362***a* of the coil pair **360***a* and the second coil **362***b* of the coil pair **360***b* are simultaneously wound around. Therefore, it is favorable for reducing the manufacturing process and lowering the cycle time of the product.

FIG. **3**E is an object-side schematic view of the coil pairs **360***a*, **360***b* according to the 3rd example in FIG. **3**A. FIG. **3**F is an image-side schematic view of the coil pairs **360***a*, **360***b* according to the 3rd example in FIG. **3**A. FIGS. **3**G to **3**I are side schematic views of the coil pairs **360***a*, **360***b* according to the 3rd example in FIG. **3**A. In FIGS. **3**A to **3**I, the top layer coil has two wire terminals, and the bottom layer coil has two wire ends, wherein the coil pair can further include a connecting wire connected to the wire terminal of the top layer coil and the wire end of the bottom layer coil so as to keep the coil pair electrically connected. According to the 3rd example, the bottom layer coil **363***a* of the first coil **361***a* of the coil pair **360***a* has a wire end **367***a*, the top layer coil **364***a* of the first coil **361***a* has a wire terminal **365***a*, the bottom layer coil **363***c* of the second coil **362***a* of the coil pair **360***a* has a wire end **367***c*, and the top layer coil **364***c* of the second coil **362***a* has a wire terminal **365***c*, wherein the coil pair **360***a* can further include a connecting wire **366***a* connected to the wire terminal **365***a* of the top layer coil **364***a* of the first coil **361***a* and the wire end **367***c* of the bottom layer coil **363***c* of the second coil **362***a* so as to keep the coil pair **360***a* electrically connected; the bottom layer coil **363***b* of the first coil **361***b* of the coil pair **360***b* has a wire end **367***b*, the top layer coil **364***b* of the first coil **361***b* has a wire terminal **365***b*, the bottom layer coil of the second coil **362***b* of the coil pair **360***b* has a wire end (its reference numeral is omitted), and the top layer coil **364***d* of the second coil **362***d* has a wire terminal **365***d*, wherein the coil pair **360***b* can further include a connecting wire **366***b* connected to the wire terminal **365***b* of the top layer coil **364***b* of the first coil **361***b* and the wire end of the bottom layer coil of the second coil **362***b* so as to keep the coil pair **360***b* electrically connected. In particular, each of the connecting wires **366***a*, **366***b* can be a portion of each of the coil pairs **360***a*, **360***b*, but the present disclosure is not limited thereto. Therefore, the feasibility of each of the coil pairs **360***a*, **360***b* composed of the wire which is continuous can be provided.

Furthermore, the required tension of the wire for fixing the coil pair **360***a* is provided by the wire terminal **365***a* of the top layer coil **364***a* and the wire terminal **365***c* of the top layer coil **364***c* winding around the columnar structures (not shown) of the carrier element, and the required tension of the wire for fixing the coil pair **360***b* is provided by the wire terminal **365***b* of the top layer coil **364***b* and the wire terminal **365***d* of the top layer coil **364***d* winding around the columnar structures of the carrier element. Therefore, the wire can be prevented from loosing from the carrier element to promote the yield rate.

According to the 3rd example, the coil pair includes two coil pairs and four wire terminals, but the present disclosure is not limited thereto.

Further, all of other structures and dispositions according to the 3rd example are the same as the structures and the dispositions according to the 1st example and the 2nd example, and will not be described again herein.

4th Example

FIG. **4**A is a schematic view of an electronic device **40** according to the 4th example of the present disclosure. FIG. **4**B is another schematic view of the electronic device **40** according to the 4th example in FIG. **4**A. In FIGS. **4**A and **4**B, the electronic device **40** is a smart phone, and includes an imaging lens driving module **401** (as shown in FIG. **4**C), and the imaging lens driving module **401** includes a ultra-wide angle camera module **402**, a high resolution camera module **403** and a telephoto camera module **404**, wherein the imaging lens driving module **401** includes an imaging lens set (not shown), a carrier element (not shown) and a driving mechanism (not shown). Furthermore, the imaging lens driving module **401** can be one of the imaging lens driving modules according to the aforementioned 1st example to the 3rd example, but the present disclosure is not limited thereto. Therefore, it is favorable for satisfying the requirements of the mass production and the appearance of the imaging lens driving modules mounted on the electronic devices according to the current marketplace of the electronic device.

Moreover, users enter a shooting mode via the user interface **405** of the electronic device **40**, wherein the user interface **405** according to the 4th example can be a touch screen for displaying the scene and have the touch function, and the shooting angle can be manually adjusted to switch the ultra-wide angle camera module **402**, the high resolution camera module **403** and the telephoto camera module **404**. At this moment, the imaging light is gathered on an image sensor (not shown) via the imaging lens set of the imaging lens driving module **401**, and an electronic signal about an image is output to an image signal processor (ISP) **406**.

FIG. **4**C is a block diagram of the electronic device **40** according to the 4th example in FIG. **4**A. In FIGS. **4**B and **4**C, to meet a specification of the electronic device **40**, the electronic device **40** can further include an optical anti-shake mechanism **407**. Furthermore, the electronic device **40** can further include at least one focusing assisting module **410** and at least one sensing element **408**. The focusing assisting module **410** can be a flash module **409** for compensating a color temperature, an infrared distance measurement component, a laser focus module, etc. The sensing element **408** can have functions for sensing physical momentum and kinetic energy, such as an accelerator, a gyroscope, a Hall Effect Element, to sense shaking or jitters applied by hands of the user or external environments. Accordingly, the electronic device **40** equipped with an auto-focusing mechanism and the optical anti-shake mechanism **407** can be enhanced to achieve the superior image quality. Furthermore, the electronic device **40** according to the present disclosure can have a capturing function with multiple modes, such as taking optimized selfies, high dynamic range (HDR) under a low light condition, 4K resolution recording, etc. Furthermore, the users can visually see a captured image of the camera through the user interface **405** and manually operate the view finding range on the user interface **405** to achieve the autofocus function of what you see is what you get.

Moreover, the imaging lens driving module **401**, the image sensor, the optical anti-shake mechanism **407**, the sensing element **408** and the focusing assisting module **410** can be disposed on a flexible printed circuit board (FPC) (its reference numeral is omitted) and electrically connected to the associated components, such as the imaging signal processor **406**, via a connector (not shown) to perform a capturing process. Since the current electronic devices, such as smart phones, have a tendency of being compact, the way of firstly disposing the imaging lens driving module and related components on the flexible printed circuit board and secondly integrating the circuit thereof into the main board of the electronic device via the connector can satisfy the requirements of the mechanical design and the circuit layout of the limited space inside the electronic device, and obtain more margins. The autofocus function of the imaging lens driving module can also be controlled more flexibly via the touch screen of the electronic device. According to the 4th embodiment, the electronic device **40** includes a plurality of sensing elements **408** and a plurality of focusing assisting modules **410**. The sensing elements **408** and the focusing assisting modules **410** are disposed on the flexible printed circuit board and at least one other flexible printed circuit board (not shown) and electrically connected to the associated components, such as the image signal processor **406**, via corresponding connectors to perform the capturing process. In other embodiments (not shown herein), the sensing elements and the focusing assisting modules can also be disposed on the main board of the electronic device or carrier boards of other types according to requirements of the mechanical design and the circuit layout.

Furthermore, the electronic device **40** can further include, but not be limited to, a display, a control unit, a storage unit, a random access memory (RAM), a read-only memory (ROM), or the combination thereof.

FIG. **4**D is a schematic view of an image shot via the ultra-wide angle camera module **402** according to the 4th example in FIG. **4**A. In FIG. **4**D, the larger range of the image can be captured via the ultra-wide angle camera module **402**, and the ultra-wide angle camera module **402** has the function of accommodating more wide range of the scene.

FIG. **4**E is a schematic view of an image shot via the high resolution camera module **403** according to the 4th example in FIG. **4**A. In FIG. **4**E, the image of the certain range with the high resolution can be captured via the high resolution camera module **403**, and the high resolution camera module **403** has the function of the high resolution and the low deformation.

FIG. **4**F is a schematic view of an image shot via the telephoto camera module **404** according to the 4th example in FIG. **4**A. In FIG. **4**F, the telephoto camera module **404** has the enlarging function of the high magnification, and the distant image can be captured and enlarged with high magnification via the telephoto camera module **404**.

In FIGS. **4**D to **4**F, the zooming function can be obtained via the electronic device **40**, when the scene is captured via the imaging lens driving module **401** with different focal lengths cooperated with the function of image processing.

The foregoing description, for purpose of explanation, has been described with reference to specific examples. It is to be noted that Tables show different data of the different examples; however, the data of the different examples are obtained from experiments. The examples were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various examples with various modifications as are suited to the particular use contemplated. The examples depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An imaging lens driving module, comprising:

an imaging lens set having an optical axis;

a carrier element configured to dispose the imaging lens set, and comprising:

two pairs of assembling structures disposed on an outer surface of the carrier element, and extending along a direction away from the optical axis; and

a driving mechanism configured to drive the carrier element to move along a direction parallel to the optical axis, and comprising:

two coil pairs disposed on the two pairs of assembling structures of the carrier element, and comprising:

a first coil pair disposed on one of the two pairs of assembling structures of the carrier element; and

a second coil pair disposed on the other one of the two pairs of assembling structures of the carrier element, the second coil pair corresponding to the first coil pair, and each of the first coil pair and the second coil pair comprising:

a bottom layer coil wound around and directly contacted with the two pairs of assembling structures; and

a top layer coil stacked on and wound around the bottom layer coil, the top layer coil farther away from the two pairs of assembling structures than the bottom layer coil away from the two pairs of assembling structures, and the top layer coil overlapping the bottom layer coil along the direction parallel to the optical axis;

at least two magnets corresponding to the two coil pair, respectively; and

at least one elastic element coupled with the carrier element;

wherein a coiling direction of the first coil pair and a coiling direction of the second coil pair are the same during observing from the first coil pair towards the second coil pair;

wherein the at least one elastic element comprises a lower elastic element, and the lower elastic element comprises four elastic sheets electrically separated from each other;

wherein each of the first coil pair and the second coil pair of the driving mechanism is a wire composed by simultaneously winding around each of the two pairs of assembling structures of the carrier element from the bottom layer coil towards two wire terminals of the top layer coil and formed symmetrically.

2. The imaging lens driving module of claim **1**, wherein the first coil pair and the second coil pair are composed of two wires, so as to keep the first coil pair and the second coil pair electrically separated.

3. The imaging lens driving module of claim **2**, wherein the lower elastic element disposed on an image side of the imaging lens set.

4. The imaging lens driving module of claim **1**, wherein the elastic sheets are electrically connected to a wire terminal of the first coil pair and a wire end of the first coil pair and a wire terminal of the second coil pair and a wire end of the second coil pair, respectively.

5. An electronic device, comprising:

the imaging lens driving module of claim **1**.

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